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Tetherless Collaborative Transportation in Space via Velocity-Matched Pushing

Utilizing STL to enable gentle transport of objects in space using
collaborative untethered agents

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**Utilizing STL to enable gentle transport of objects
in space using collaborative untethered agents**

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Master's Programme, Systems, Control and Robotics, 120 credits
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Acknowledgments

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I would like to thank xxxx for having yyyy. Or in the case of two authors: We would like to thank xxxx for having yyyy.

Stockholm, January 2026

Arian Kourangi Esfahani

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List of acronyms and abbreviations

CBF	Control Barrier Function
KTH	KTH Royal Institute of Technology
MILP	Mixed Integer Linear Programming
MIP	Mixed Integer Programming
MIQP	Mixed Integer Quadratic Programming
MPC	Model Predictive Control
QP	Quadratic Programming
RoI	Region of Interest
STL	Signal Temporal Logic

Chapter 1

Introduction

1.1 Background

Transportation of passive objects or non-cooperative agents in the micro-gravity of space is the subject of many scientific endeavours. In-space transportation is a fundamental capability for emerging space activities such as satellite servicing, debris removal, and assembly of large infrastructure like solar power stations and telescopes [1, 2]. Unlike terrestrial transport, there are no natural stabilizing forces such as gravity or friction. Once an object is set in motion, it will continue indefinitely, making even simple manoeuvres highly non-trivial.

A large body of work exists on transporting non-cooperative objects in microgravity, ranging from single-robot to multi-agent solutions, and using various control schemes and collaboration methods to transport objects rigidly anchored to the robots [3, 4]. Furthermore, to facilitate transport when robots are not physically connected to objects, several studies focus on trajectory planning and impact kinematics for capturing objects with manipulators [5, 6]. [7] reviews multiple strategies for on-orbit service, highlighting the need for careful planning and control during robot-object interactions to ensure safe and reliable docking or capturing.

Collaborative transport is particularly attractive because it allows multiple small robots to replace a single heavy manipulator system [8], which is typically more costly to deploy and performs slower than collaborative systems. However, coordinating multiple free-flying agents introduces challenges such as coupled dynamics, synchronization of contact forces, and stability of the object's trajectory [9].

As noted in the aforementioned studies, most transportation methods

in microgravity rely on rigid connections, tethers, or manipulators which require complex trajectory planning in addition to position and velocity consensus between robots and objects to facilitate capture or docking before transport. In a more recent study, [10] demonstrated a method for collaborative transportation via impact interactions in microgravity, utilizing **Signal Temporal Logic (STL)** and **Mixed Integer Linear Programming (MILP)** to formalize constraints and interactions between robots and objects to achieve the desired transport. This method eliminates the need for rigid anchoring, reducing system complexity and deployment costs.

Using impacts in this way introduces other concerns, particularly the risk of damage to the robot or object during collision. It also increases the danger of contributing to the growing problem of space debris. Another issue is the inherent under-actuation of the object. Without a rigid connection, slipping can occur, and controlling orientation or rotational velocity becomes very difficult. This lack of full control authority not only reduces precision but also increases the likelihood of unstable or unsafe trajectories.

In summary, while impact-based approaches enable anchor-free collaborative transport, they remain limited by the inability to ensure safe, compliant, and controlled interactions.

1.2 Problem

For the long-term viability of anchor-free collaborative transport, a shift towards compliant, soft-contact interactions is required. In the space domain, this problem is closely related to the well-studied challenge of rendezvous, where two objects, such as satellites, must achieve simultaneous consensus in both position and velocity before docking or interaction can occur. Unlike terrestrial or aerial robotics, rendezvous in microgravity is particularly demanding. Objects may tumble unpredictably, relative motion must be estimated under significant sensing uncertainty, and even minor mismatches can lead to destabilization or mission failure [7]. The study of automated control strategies for rendezvous has therefore been the focus of a large body of research. For example, [11] provides a comprehensive overview of classical and modern approaches, highlighting the persistent challenges of robustness, safety, and efficiency. Typically, the latter phase of these approaches relies on feedback control laws that use online estimations of the target's pose and velocity to track the planned trajectory and achieve rendezvous. Since the motion of the target is highly uncertain, high-level planning over large horizons remains an area of open research, made difficult by the combined effects of

non-linear orbital dynamics, uncertain target motion, and tight operational constraints.

Interaction or transport of objects without rigid connections further complicates control. Since the object is not physically constrained, forces can only be applied through a single contact point during interaction. Consequently, the direction of the applied force must remain approximately constant throughout contact, severely restricting the set of feasible maneuvers and making it difficult to control both the translational and rotational motion of the object simultaneously. Therefore, achieving precise and stable transport under these conditions requires coordination and planning that explicitly consider these physical limitations and constraints.

Given these physical and control challenges, a structured way to encode and reason about the constraints of such interactions is required. Formal specification methods such as **STL** are particularly well-suited for this domain. **STL** allows encoding of high-level temporal and safety constraints into mathematically precise conditions [12]. This provides a systematic way to design controllers that guarantee safety and correctness, which is crucial in space missions where failure has irrecoverable consequences. However, **STL** synthesis faces well-known challenges in scalability with respect to task complexity and planning horizon [13]. Consequently, much of the recent scientific effort has focused on extending **STL** to handle more complex task descriptions, multi-agent collaboration, and online execution. For example, Lindemann *et al.* [14, 15] propose a framework combining time-varying control barrier functions with **STL** to ensure satisfaction of complex tasks and cooperative behaviours among multiple agents. Similarly, Zhao *et al.* [16] apply such methods to enable safe merging manoeuvres in autonomous vehicles.

In summary, the problem addressed in this thesis lies in the absence of a formalized framework for planning and controlling compliant, anchor-free transport in microgravity. Existing methods either rely on rigid anchoring or impulsive impacts, both of which introduce limitations in precision, safety, and reusability. By leveraging formal methods such as **STL** to encode the temporal and physical constraints inherent in soft-contact interactions, this thesis aims to establish a foundation for safe, reliable, and scalable anchor-free collaborative transport in space.

1.2.1 Research Question

How can rendezvous-preceded interactions be formally specified using **STL** to enable gentle, anchor-free collaborative transport of a free-floating, non-cooperative object in microgravity?

1.3 Purpose

The purpose of this thesis is to design and implement a software and control framework for anchor-free collaborative transport of passive objects in microgravity. Building upon the impact-based transport approach of [10], this work replaces impulsive interactions with compliant, rendezvous-based contacts. The proposed method formally specifies rendezvous requirements and interaction stages using **STL** and **Mixed Integer Programming (MIP)**, integrating these into a hierarchical motion planning and control architecture. In doing so, the thesis preserves the benefits of anchor-free transport while improving safety, stability, and precision.

1.4 Goals

To achieve the stated purpose, the goals of this thesis include:

1. Developing a high-level planner capable of specifying rendezvous preceded robot-object interactions and high-level task specification using **STL** and **MIP**.
2. Developing a control strategy that allows the robots to interact with the object in four distinct phases.
 - (a) Rendezvous and Approach: Synchronize the velocity and position of the robots with the object and execute a controlled approach trajectory.
 - (b) Non-grasping Contact: Establish safe contact with the object without mechanical grasping or interlocking, ensuring stability while minimizing risks of slippage or induced rotation.
 - (c) Velocity Transfer: Impart a desired translational velocity to the object.
 - (d) Separation and Stop: Disengage from the object, ensuring that the robots return to a stable state without imparting unintended forces or torques on the object.

3. Evaluate the developed methods in two-dimensional microgravity-analog environments. This includes both physics-based simulations of robots on air bearings and experiments on a physical testbed with real robots operating on air-bearing platforms.
4. Ensure that the developed algorithms handle uncertainties such as modeling errors, sensor noise, or small disturbances, while maintaining safety and preventing object destabilization.

1.5 Research Methodology

The methodology used in this thesis largely follows the Design Science Research Process as presented in [17]:

1. Problem Identification and Motivation.
2. Defining the objectives of a solution.
3. Design and development.
4. Demonstration.
5. Evaluation.
6. Communication.

As described in Chapter 3, the method developed in this study is comprised of multiple planners and controllers to achieve the desired goal. Each planner or controller was developed separately in an iterative fashion. Each iteration involved implementing a partial design, running simulations to evaluate key aspects such as computation time, feasibility of the desired trajectory and interaction, and adjusting the design based on the observed limitations.

1.6 Delimitations

- No experiments will be conducted in actual or simulated three-dimensional microgravity. Instead, the evaluation will be restricted to two-dimensional analog environments, where free-floating objects are supported by air-bearing platforms.
- Both robots and objects will be modelled with simplified circular geometries, and all interactions will be constrained to a 2D plane.

- The primary emphasis will be placed on simulation studies in 2D environments. Physical experiments on the air-bearing test-bed will only be pursued once sufficient progress has been achieved in simulation.
- Ground-truth motion data of robots and objects will be used for control and evaluation purposes, obtained either directly from simulation software or through motion-capture systems in experiments. On-board state estimation methods (e.g. perception) will not be developed, as this would significantly expand the scope of the thesis.
- The interaction framework will not consider highly irregular object geometries, deformable bodies, variable mass distributions, or matching of rotational velocities, as these add substantial modelling and control complexity that exceeds the scope of the project.

1.7 Structure of the thesis

Rapportens disposition

Chapter 2 presents relevant background information about xxx. Chapter 3 presents the methodology and method used to solve the problem. ...

Chapter 2

Background

2.1 Related Work

This thesis builds on and extends the framework introduced by [10], who first demonstrated a method for collaborative transport in microgravity using impact-based interaction. Their formulation and methodological insights form the foundation for several components of the approach developed in this thesis. In particular, the treatment of trajectory parametrization and the encoding of **STL** specifications within **MILP** closely follows the structure established in their work.

While the theoretical underpinnings and overall problem formulation draw heavily from [10], this thesis advances the framework by adapting it to scenarios where collaborative manoeuvres rely on rendezvous-based pre-contact interactions rather than direct impacts. This shift enables safer and less aggressive transport behaviours, which are better suited for delicate payloads and constrained operational environments. The resulting extensions generalize the original method and broaden its applicability, demonstrating how the underlying principles of collaborative transport can be maintained while significantly altering the interaction modality.

2.2 Signal Temporal Logic

STL [12] is widely used in cyber-physical and dynamical systems as a formal language for specifying real-time requirements such as safety constraints, temporal objectives, and conditional system behaviours. An **STL** formula is

built from atomic predicates of the form

$$\mu := f(\mathbf{s}(t)) \geq 0,$$

where f is typically a linear function that evaluates some property of the real-valued signal $\mathbf{s}(t)$ at time t .

The syntax of **STL** is defined recursively as

$$\varphi ::= \mu \mid \neg\varphi \mid \varphi_1 \wedge \varphi_2 \mid \varphi_1 \mathbf{U}_{[a,b]} \varphi_2, \quad (2.1)$$

where $\neg$ and $\wedge$ denote Boolean negation and conjunction, and $\mathbf{U}_{[a,b]}$ is the time-bounded *Until* operator. The formula $\varphi_1 \mathbf{U}_{[a,b]} \varphi_2$ states that φ_1 must hold continuously until φ_2 becomes true at some time within the interval $[a, b]$.

Using the *Until* operator, additional temporal operators can be defined. The *Eventually* operator $\mathbf{F}_{[a,b]}\varphi$ requires that φ holds at some time in the interval $[a, b]$, and the *Globally* operator $\mathbf{G}_{[a,b]}\varphi$ requires that φ holds at all times in $[a, b]$.

A summary of the Boolean semantics of **STL** is provided in equations 2.2-2.7.

$$(\mathbf{s}, t) \models \mu \Leftrightarrow f(\mathbf{s}(t)) \geq 0, \quad (2.2)$$

$$(\mathbf{s}, t) \models \neg\varphi \Leftrightarrow (\mathbf{s}, t) \not\models \varphi, \quad (2.3)$$

$$(\mathbf{s}, t) \models \varphi_1 \wedge \varphi_2 \Leftrightarrow (\mathbf{s}, t) \models \varphi_1 \text{ and } (\mathbf{s}, t) \models \varphi_2, \quad (2.4)$$

$$(\mathbf{s}, t) \models \mathbf{F}_{[a,b]}\varphi \Leftrightarrow \exists t' \in [t + a, t + b] : (\mathbf{s}, t') \models \varphi, \quad (2.5)$$

$$(\mathbf{s}, t) \models \mathbf{G}_{[a,b]}\varphi \Leftrightarrow \forall t' \in [t + a, t + b] : (\mathbf{s}, t') \models \varphi, \quad (2.6)$$

$$(\mathbf{s}, t) \models \varphi_1 \mathbf{U}_{[a,b]} \varphi_2 \Leftrightarrow \exists t' \in [t + a, t + b] : (\mathbf{s}, t') \models \varphi_2 \quad (2.7) \\ \wedge \forall \tau \in [t, t'] : (\mathbf{s}, \tau) \models \varphi_1.$$

To support optimization-based testing and trajectory synthesis, STL can also be equipped with a *robustness* value $\rho(\mathbf{s}, t, \varphi)$, which quantifies not only whether a signal satisfies a formula but by what margin. Robust semantics assign a positive value when the formula is satisfied, a negative value when it is violated, and a magnitude that reflects the distance to the satisfaction boundary. This continuous measure enables STL specifications to be incorporated directly into numerical optimization frameworks, including **MILP**, nonlinear programming, and sampling-based search methods, where

purely Boolean semantics would provide no gradient or direction of improvement. Equations 2.8–2.11 illustrate some representative robustness measures. For a complete treatment of STL semantics we refer the reader to [12, 18].

$$\rho(\mathbf{s}, t, \varphi) > 0 \Leftrightarrow (\mathbf{s}, t) \models \varphi, \quad (2.8)$$

$$\rho(\mathbf{s}, t, \varphi) < 0 \Leftrightarrow (\mathbf{s}, t) \not\models \varphi, \quad (2.9)$$

$$\rho(\mathbf{s}, t, \mathbf{G}_{[a,b]}\varphi) = \min_{t' \in [t+a, t+b]} \rho(\mathbf{s}, t', \varphi) \quad (2.10)$$

$$\rho(\mathbf{s}, t, \mathbf{F}_{[a,b]}\varphi) = \max_{t' \in [t+a, t+b]} \rho(\mathbf{s}, t', \varphi) \quad (2.11)$$

2.3 Mixed Integer Programming

2.3.1 Introduction and Motivation

MIP are optimization frameworks commonly used in decision making, planning, and control. What makes this framework particularly powerful is its ability to combine a *mixture* of decision variables, most notably continuous variables and binary or integer variables, within a single model. This is especially useful for dynamic systems that feature both continuous quantities (e.g., positions, velocities, energies) and discrete decisions (e.g., on/off, left/right, activate/deactivate). By allowing these two types of variables to coexist, MIP enables the synthesis of plans, trajectories, and control strategies that simultaneously satisfy dynamical, physical, and logical constraints while still pursuing a specified objective.

In the context of robotics and control, a MIP model is typically constructed by defining decision variables and imposing constraints that encode system dynamics, kinematics, safety requirements, and logical relations. An objective function then guides the optimizer toward a desirable solution, such as minimizing time, maximizing safety margins, or achieving a goal state efficiently. The exact structure of the model determines the terminology used: if all constraints and the objective are linear, the model is referred to as a *Mixed-Integer Linear Program (MILP)*; if quadratic terms are used in the objective, it becomes a *Mixed Integer Quadratic Programming (MIQP)*, and so on.

For the purposes of this thesis, a deep mathematical treatment of **MIP** is

not required. The underlying theory is rigorous and broad, and an extensive exposition would detract from the focus of the work. Therefore, only the essential modelling concepts are introduced here, while an interested reader is referred to [19–21] for a more comprehensive background.

2.3.2 Modelling Primitives in MILP

A practical challenge when working with **MILP** models is that even relatively simple constraints can become notationally cumbersome when expressed explicitly in linear inequality form. To avoid interrupting the flow of later chapters with low-level formulations, this background section introduces the standard modelling primitives that will be used throughout the thesis. The examples below illustrate how system dynamics, logical relations, and hybrid behaviours can be represented using linear constraints and mixed-integer decision variables.

2.3.2.1 Linear Equality Constraints

Many dynamical systems evolve according to linear equations of the form

$$x_{t+1} = Ax_t + Bu_t, \quad (2.12)$$

which translate directly into MILP constraints:

$$x_{t+1} - Ax_t - Bu_t = 0, \quad t = 0, \dots, T - 1. \quad (2.13)$$

Linear equalities of this form are used to encode deterministic system dynamics, kinematic relations, and other structural dependencies.

2.3.2.2 Linear Inequality Constraints

Physical and safety bounds are typically expressed as linear inequalities, such as

$$x_{\min} \leq x_t \leq x_{\max}, \quad u_{\min} \leq u_t \leq u_{\max}. \quad (2.14)$$

These can be written in standard MILP form as

$$x_t - x_{\max} \leq 0, \quad x_{\min} - x_t \leq 0, \quad (2.15)$$

and analogously for the input u_t . Such constraints enforce workspace limits, actuator capabilities, and admissible operating regions.

2.3.2.3 Polyhedral Region Membership

Atomic conditions such as remaining inside a safe region or satisfying a state-dependent predicate are represented as halfspace constraints of the form

$$a^\top x_t \leq b. \quad (2.16)$$

2.3.2.4 Logical Implication via Big- M

Binary variables enable the encoding of logical implications. A statement of the form

$$z_t = 1 \Rightarrow a^\top x_t \leq b \quad (2.17)$$

is enforced using the standard Big- M relaxation:

$$a^\top x_t \leq b + M(1 - z_t), \quad (2.18)$$

where M is chosen large enough so that the constraint becomes inactive when $z_t = 0$. Such constructions form the backbone of MILP-based encodings of Boolean STL operators.

2.3.2.5 Disjunction (OR) Constraints

To represent a logical disjunction such as

$$a^\top x_t \leq b \quad \text{or} \quad c^\top x_t \leq d, \quad (2.19)$$

we introduce two binary variables $z_{1,t}, z_{2,t}$ and enforce

$$z_{1,t} + z_{2,t} = 1, \quad (2.20)$$

$$a^\top x_t \leq b + M(1 - z_{1,t}), \quad (2.21)$$

$$c^\top x_t \leq d + M(1 - z_{2,t}). \quad (2.22)$$

At most one constraint must hold tightly, ensuring that at least one of the corresponding regions is selected.

2.3.2.6 Binary Activation of Equality Constraints

Sometimes a binary variable must enforce that a particular equality holds *only when activated*. For a constraint of the form

$$z_t = 1 \Rightarrow x_t = c, \quad (2.23)$$

we use both upper and lower Big- M bounds:

$$x_t \leq c + M(1 - z_t), \quad (2.24)$$

$$x_t \geq c - M(1 - z_t). \quad (2.25)$$

When $z_t = 1$, both inequalities reduce to $x_t = c$. When $z_t = 0$, the bounds become inactive and x_t is unconstrained within the range implied by M .

2.4 Bézier Curves

Bézier curves are a family of parametric curves defined by polynomial functions and a finite set of control points. Widely used in computer graphics and geometric modeling, they have also gained increasing attention in motion planning. Their smooth polynomial structure makes them particularly suitable for trajectory generation [22, 23].

A Bézier curve of degree n is defined by a sequence of $n + 1$ control points

$$\mathbf{P}_0, \mathbf{P}_1, \dots, \mathbf{P}_n \in \mathbb{R}^d,$$

where d is the dimension of the space (typically $d = 2$ or 3). The *degree* of the curve determines the order of the underlying polynomial and governs the curve's expressiveness: higher-degree curves allow for more complex shapes but also require more control points and computational effort.

The Bézier curve is parameterized over the phasing parameter $s \in [0, 1]$ and expressed as a weighted sum of its control points using the Bernstein basis polynomials:

$$\mathbf{B}(s) = \sum_{i=0}^n \mathbf{P}_i b_{i,n}(s),$$

where the Bernstein polynomials of degree n are given by

$$b_{i,n}(s) = \binom{n}{i} (1-s)^{n-i} s^i.$$

The control points act as geometric handles that shape the curve. While the curve does not in general pass through all control points (except for the first and last), it always remains within the convex hull defined by the control points, a property particularly valuable in trajectory planning where geometric guarantees on feasibility are important. Moreover, the first and last control points determine the start and end of the curve and the curves derivative is itself a Bézier curve of one degree lower, obtained as linear combinations of the original control points:

$$\frac{d\mathbf{B}(s)}{ds} = n \sum_{i=0}^{n-1} (\mathbf{P}_{i+1} - \mathbf{P}_i) b_{i,n-1}(s),$$

which shows that the derivative curve is defined by the n control points

$$\mathbf{P}'_i = n(\mathbf{P}_{i+1} - \mathbf{P}_i), \quad i = 0, \dots, n-1.$$

2.4.1 Parametrizing Trajectories

Parametrizing physical trajectories using Bézier curves offers significant advantages over fine-grained discretization, as it requires far fewer parameters to represent smooth curves. This reduction in dimensionality directly decreases computational complexity in motion planning and trajectory optimization problems. While a purely spatial parametrization can be achieved using a single Bézier curve, such an approach ignores the temporal structure of the trajectory, which is critical for satisfying dynamic constraints.

To address this, [22] proposes a method for parametrizing trajectories over time using two Bézier curves: a *spatial* curve $r(s)$ and a *temporal* curve $h(s)$, both defined over the phasing parameter $s \in [0, 1]$. The actual physical trajectory $p(t)$ is then expressed by composing these two curves. Specifically, the trajectory evaluated at s is given by

$$r(s) = p(h(s)), \quad (2.26)$$

and, by the chain rule, its derivative satisfies

$$\dot{r}(s) = \dot{p}(h(s)) \dot{h}(s). \quad (2.27)$$

This parametrization separates the geometric shape of the trajectory from its temporal evolution, enabling independent control of spatial and timing characteristics.

Chapter 3

Method

3.1 Preface

It is important to note that several elements of this framework are adapted from [10], whose formulation provides the basis for the overall hierarchy. In particular, the structure of the pipeline and the approach taken in Sections 3.1.2, 3.2.1, 3.2.2, 3.2.3.4 and 3.2.3.5 closely follow their original presentation. These components are retained here due to their effectiveness and relevance to the problem setting, while the remaining sections introduce the novel extensions and modifications developed in this thesis.

3.1.1 Problem Formulation

Consider the set of robots $\mathcal{R} = \{R_1, \dots, R_{|\mathcal{R}|}\}$ and the set of objects $\mathcal{O} = \{O_1, \dots, O_{|\mathcal{O}|}\}$, and define $\mathcal{S} = \mathcal{R} \cup \mathcal{O}$ as the set of all systems involved in the task. Each system $S \in \mathcal{S}$ is described by the state vector $x_S = [p_S, q_S, \dot{p}_S, \dot{q}_S]$, where p_S and q_S denote its position and orientation, and $\dot{p}_S$ and $\dot{q}_S$ their respective time derivatives.

Because the solution must explicitly determine *when* and *between whom* interactions occur, we introduce the binary interaction variable $interact(S_i, S_j, t) \in \{\perp, \top\}$, which indicates whether system S_i is interacting with system S_j at time t .

We can now formulate the problem for collaborative transport using rendezvous preceded interactions as follows:

$$\min_{x_{R_i}, u_{R_i}, x_{O_j}} J(\mathbf{x}_{\mathcal{R}}, \mathbf{u}_{\mathcal{R}}, \mathbf{x}_{\mathcal{O}}), \quad (3.1)$$

subject to

$$[\mathbf{x}_{\mathcal{R}}, \mathbf{x}_{\mathcal{O}}] \models \varphi, \quad (3.2)$$

$$\dot{\mathbf{x}}_{\mathcal{R}} \in \mathcal{V}_{\mathcal{R}}, \dot{\mathbf{x}}_{\mathcal{O}} \in \mathcal{V}_{\mathcal{O}}, \quad (3.3)$$

$$\dot{x}_{R_i} = \begin{bmatrix} \dot{p}_{R_i} & \dot{q}_{R_i} & u_{R_i} \end{bmatrix}^T \quad (3.4)$$

$$\dot{x}_{O_i} = \begin{cases} \begin{bmatrix} \dot{p}_{R_j} & \dot{q}_{R_j} & 0 \end{bmatrix}^T & \text{if } \text{interact}(O_i, R_j, t) \\ \begin{bmatrix} \dot{p}_{O_i} & \dot{q}_{O_i} & 0 \end{bmatrix}^T & \text{otherwise} \end{cases} \quad (3.5)$$

$$\forall i, j \in \mathcal{S}, i \neq j, \forall t \in [t_0, t_f] : \quad (3.6)$$

$$\neg \text{interact}(S_i, S_j, t) \iff \|p_{S_i}(t) - p_{S_j}(t)\|_2 \geq \text{rad}_{S_i} + \text{rad}_{S_j} \\ \forall i, j \in \mathcal{S}, i \neq j, \forall t \in [t_0, t_f] : \quad (3.7)$$

$$\text{interact}(S_i, S_j, t) \iff \|p_{S_i}(t) - p_{S_j}(t)\|_2 = \text{rad}_{S_i} + \text{rad}_{S_j}$$

where $\mathbf{x}_{\mathcal{R}}$ and $\mathbf{x}_{\mathcal{O}}$ denote the aggregated states of all robots and objects, respectively, and $\mathcal{V}_{\mathcal{R}}$ and $\mathcal{V}_{\mathcal{O}}$ represent the permissible velocity bounds for each group. The state of an individual system $S_i \in \{R, O\}$ is given by $x_{S_i} = [p, q, \dot{p}, \dot{q}]$, u_{R_i} denotes the control input applied to robot R_i and rad its radius. The term φ captures the **STL** specifications, including workspace limits, transportation objectives, collision avoidance, and obstacle avoidance.

Equations 3.2–3.3 ensure that all system trajectories satisfy the high-level STL specifications and remain within the allowable velocity bounds. Equations 3.4–3.5 enforce that each robot’s state is governed by its own input, while objects adopt the state of the interacting robot during an interaction. Equation 3.6 ensures that the distance between any two non-interacting systems remains larger than the sum of their radii. Finally, Equation 3.7 specifies the required conditions for the transfer of normal forces between two systems during an interaction.

3.1.2 Approach

Following the hierarchical framework presented in [10], our method is divided into three main components. First, an offline planner that operates on simplified kinematic models, treating all robots and objects as point masses. This planner generates nominal trajectories and determines when, where, and with whom robots and objects should interact. Second, an online re-planner incorporates the physical dimensions of all systems and enforces feasibility

constraints on trajectories and interactions. It also continuously updates plans based on the current estimated states of all robots and objects. Finally, a **Model Predictive Control (MPC)** ensures accurate tracking of the generated trajectories and the successful execution of interactions.

3.2 Offline Planner

3.2.1 Introduction

For the synthesis of a plan, it is necessary to determine how the collaborative transport task should be executed in order to best satisfy the overall objective. At the same time, the generated motions must resemble physically feasible trajectories and interactions as closely as possible. To meet these requirements, we employ a **MILP** planner. Due to the computational complexity associated with solving such problems, the **MILP** planning is performed offline.

To further reduce computational demands, both robots and objects are modelled as point masses, and their orientations and rotational velocities are omitted from the planning formulation. Additionally, directly reasoning over continuous trajectory representations in the workspace would severely hinder **MILP** performance, as it would require parametrizing robot trajectories with an intractably large number of variables. To address this, and following the approach in [10], detailed in Section 2.4.1, we parametrize robot and object trajectories using Bézier curves. This representation significantly reduces the number of decision variables required for the planner to operate, enabling efficient computation of feasible trajectories. As demonstrated in [24], this approach provides substantial computational benefits while preserving trajectory expressiveness.

Since objects are only to be moved through interactions with robots, their spatial ($r(s)$) and temporal ($h(s)$) curves are limited to 1st degree Bézier curves, whilst robots are permitted higher degrees.

3.2.2 MILP Encoding of STL Specifications

Using the modelling primitives introduced in Section 2.3.2, **STL** formulas can be encoded exactly as mixed-integer linear constraints. Instead of enumerating all operators exhaustively, we illustrate the procedure through representative examples and outline how these components integrate into a full **STL**-constrained optimization problem. For details the reader is referred to [25].

Example: Predicate. A predicate has the form

$$\mu(x_t) \equiv a^\top x_t \leq b.$$

To encode whether this predicate is satisfied at time t , we introduce a binary variable $z_{\mu,t}$ and impose

$$\begin{aligned} a^\top x_t &\leq b + M(1 - z_{\mu,t}), \\ a^\top x_t &\geq b - Mz_{\mu,t}. \end{aligned}$$

When $z_{\mu,t} = 1$, the constraints enforce $a^\top x_t \leq b$; when $z_{\mu,t} = 0$, the inequalities are relaxed by the Big- M terms. This pattern forms the basis for encoding atomic propositions in STL.

Example: Globally Operator. The temporal operator $\mathbf{G}_{[a,b]} \varphi$ requires that φ holds at every time $t' \in [t+a, t+b]$. Introducing binary variables $z_{\varphi,t'}$ for the satisfaction of φ at each time, we encode:

$$\begin{aligned} z_{\mathbf{G}_{[a,b]} \varphi, t} &\leq z_{\varphi, t'}, & \forall t' \in \{t+a, \dots, t+b\}, \\ z_{\mathbf{G}_{[a,b]} \varphi, t} &\geq \sum_{t'=t+a}^{t+b} z_{\varphi, t'} - (b-a). \end{aligned}$$

These constraints ensure that $z_{\mathbf{G}_{[a,b]} \varphi, t} = 1$ if and only if φ is satisfied at all times within the interval.

In general, each STL subformula φ is associated with a binary variable $z_{\varphi,t} \in \{0, 1\}$ that denotes whether the formula is satisfied at time t . Compositional STL operators (e.g., conjunctions, disjunctions, and other temporal operators) are encoded by linking these binary variables through standard logical and temporal constructions.

3.2.2.1 Integration into the MILP Objective

As discussed in Section 2.2, each atomic predicate is assigned a robustness value (e.g., space-based or time-based). Following the approach in [10, 25], these robustness values are propagated through the STL syntax tree via min and max operators. The resulting robustness at the root formula φ provides a scalar measure of spatial/temporal satisfaction. This root robustness value is then incorporated directly into the MILP as part of the objective function. We can thus formulate the spatial-robust planning problem as follows:

$$\max_{r_{R_i}, h_{R_i}, r_{O_j}, h_{O_j}} J(r_{R_i}, h_{R_i}, r_{O_j}, h_{O_j}) = \rho_\varphi(\mathbf{x}_R, \mathbf{x}_O) \quad (3.8)$$

s.t.

$$\dot{x}_{R_i} = \begin{Bmatrix} \dot{p}_{R_i} & u_{R_i} \end{Bmatrix}^T \quad (3.9)$$

$$\dot{x}_{O_i} = \begin{cases} \begin{Bmatrix} \dot{p}_{R_j} & 0 \end{Bmatrix}^T & \text{if } \text{interact}(O_i, R_j, t) \\ \begin{Bmatrix} \dot{p}_{O_i} & 0 \end{Bmatrix}^T & \text{otherwise} \end{cases} \quad (3.10)$$

$$\begin{aligned} & \forall i, j \in \mathcal{S}, i \neq j, \forall t \in [t_0, t_f] : \\ & \text{interact}(S_i, S_j, t) \iff \|p_{S_i}(t) - p_{S_j}(t)\|_2 = 0 \\ & \text{Eqs. (3.2), (3.3), (3.6)} \end{aligned} \quad (3.11)$$

3.2.3 MILP encoding of kinematic and logical constraints

This section presents the kinematic and logical constraints used to encode trajectories and interactions within the **MILP**. For clarity and readability, we omit the full **MILP** formulations and instead provide simplified representations of the constraints. Nevertheless, all constraints discussed here are implemented using the standard primitives introduced in Section 2.3.2. To support a consistent and interpretable presentation, the notation used throughout this chapter is summarized in Table 3.1. Note that, instead of allowing interactions to occur at arbitrary continuous times, we restrict interactions to take place only between a single robot Bézier curve and a single object Bézier curve at any given moment. Moreover, interactions are permitted only at the start or end of these curves. To encode the logic that activates or relaxes constraints depending on whether an interaction is present, we introduce binary variables $z_{R,O} \in \mathbb{B}^{(N_R \times N_O)}$, where N_R and N_O denote the number of Bézier curves assigned to robot R and object O , respectively. These binaries serve a role analogous to $\text{interact}(O_i, R_j, t)$, but are defined per curve pair rather than per discrete time step t .

*In practice this velocity curve does not exist. It is only a compact representation of $\frac{\dot{r}}{h}$. For the purposes of this thesis we instead use the scalar velocity variable v when writing velocity constraints, since enforcing one equality on v would otherwise require introducing two separate constraints on $\dot{r}$ and $\dot{h}$.

r	Spatial curve
h	Temporal curve (time)
v	Velocity curve ($\frac{\dot{r}}{h}$)*
r_R	Spatial curve of robot R
r_O	Spatial curve of object O
$r^{(k)}$	Curve with index k
$r^{(k)(0)}$	First control point of curve k
$r^{(k)(n)}$	Last control point of curve k
$\hat{r}$	Replanned curve
P_R^t	Measured state of robot R at time t (position $[x, y]$)
ρ_R	Radius of robot R
N_R	Number of Bézier curves assigned to robot R
$z_{(R,O)}[k, j]$	Interaction binary for curve k of robot R and curve j of object O
m_R	Mass of robot R

Table 3.1: Notation

3.2.3.1 Continuity Constraints

To ensure that the generated trajectories are kinematically feasible and do not include discrete jumps between Bézier segments we enforce the following constraints:

$$\begin{aligned}
 h_R^{(k)(n)} &= h_R^{(k+1)(0)} \\
 r_R^{(k)(n)} &= r_R^{(k+1)(0)} \quad \forall k \in N_R \\
 v_R^{(k)(n)} &= v_R^{(k+1)(0)}
 \end{aligned} \tag{3.12}$$

For reason that will be discussed, we cannot enforce continuity of the object state during interactions. To that end, we enforce continuity of object Bézier segments only for segments that are not part of an interaction:

$$\begin{aligned}
 h_O^{(j)(n)} &= h_O^{(j+1)(0)} \\
 r_O^{(j)(n)} &= r_O^{(j+1)(0)} \iff z_{(R,O)}[k, j] = \perp, \forall R \in \mathcal{R} \forall k \in N_R \\
 v_O^{(j)(n)} &= v_O^{(j+1)(0)}
 \end{aligned} \tag{3.13}$$

3.2.3.2 Interaction Constraints

For a robot and an object to engage, the interaction must be preceded by a rendezvous, meaning that their positions and velocities coincide at a specific point in time. Accordingly, we define an interaction between robot Bézier segment k and object Bézier segment j as follows:

$$\begin{aligned} h_R^{(k)(0)} &= h_O^{(j)(0)} \\ r_R^{(k)(0)} &= r_O^{(j)(0)} \iff z_{(R,O)}[k, j] = \top \\ v_R^{(k)(0)} &= v_O^{(j)(0)} \end{aligned} \quad (3.14)$$

During an interaction, the robot and object states are conceptually assumed identical. Since object curves are parametrized using fewer control points we cannot enforce this in the solver. Instead, the solver assumes the objects velocity during the interaction to be fixed at the velocity it had before the interaction. In reality the robot is manipulating its velocity during the interaction, but updating this is deferred until the next curve as follows:

$$\begin{aligned} h_R^{(k+1)(0)} &= h_O^{(j+1)(0)} \\ r_R^{(k+1)(0)} &= r_O^{(j+1)(0)} \iff z_{(R,O)}[k, j] = \top \\ v_R^{(k+1)(0)} &= v_O^{(j+1)(0)} \end{aligned} \quad (3.15)$$

To achieve this, we have to permit discontinuities in the object state between an interaction and its proceeding curve, see equations 3.13. To keep the notation tighter and interpretable, for the rest of this paper we will refer to the objects state at the end of the interaction as $X_O^{(j)(n)}$, even though in practice the update of the objects state doesn't occur until the beginning of the next curve.

3.2.3.3 Robot Interaction Constraints

Since there is no mechanical coupling between the robot and the object, the robot cannot apply force in an arbitrary direction during an interaction. Instead, for each interaction, it must commit to a specific quadrant of allowable force directions. The reason is that if the sign of the applied force were to change within an interaction, the robot would need to physically reposition itself, as forces can only be transmitted through normal contact. To model this, the robot selects the sign of the applied force in each dimension. The binary variables F^+ and F^- indicate the chosen sign of the acceleration (or

force) along a given coordinate axis. *

$$\begin{aligned}
 v_R^{(k)(i+1)} - v_R^{(k)(i)} &\geq 0 &\iff F^+ \\
 v_R^{(k)(i+1)} - v_R^{(k)(i)} &\leq 0 &\iff F^- \\
 \forall i \in [0, n-1] &\iff \\
 z_{(R,O)}[k, j] &= \top \quad \exists j \in N_O \quad \exists O \in \mathcal{O}
 \end{aligned} \tag{3.16}$$

3.2.3.4 Obstacle & Collision Avoidance

Following the method in [10], containment of a Bézier curve within a **Region of Interest (RoI)** can be enforced by constraining all of its control points to satisfy the linear inequalities defining the **RoI** polytope. Let the **RoI** be given in the half-space representation $Hx \leq b$, and let n^{faces} denote the number of faces of the polytope. Then the condition

$$x_S \in \text{RoI} \iff \left(\bigwedge_{f=1}^{n^{\text{faces}}} \bigwedge_{i=0}^d (H^f r^{(j)(i)} \leq b^f) \right), \quad \forall j \in \mathbb{N}_{[0, N-1]},$$

ensures that every control point of the i -th Bézier segment lies inside all bounding half-spaces of the **RoI**.

Conversely, collision avoidance with a **RoI** can be imposed by enforcing that at least one control point violates one of the supporting half-spaces. This can be expressed as

$$x_S \notin \text{RoI} \iff \left(\bigvee_{f=1}^{n^{\text{faces}}} \bigwedge_{i=0}^d (H^f r^{(j)(i)} \geq b^f) \right), \quad \forall j \in \mathbb{N}_{[0, N-1]},$$

where the disjunction over faces ensures that each Bézier segment remains strictly outside at least one face of the polytope.

3.2.3.5 Acceleration Attenuation

Since the offline planner determines the full kinematic and interaction structure of the control stack without considering the dynamics, ensuring

*Because $v = \frac{\dot{r}}{\dot{h}}$, encoding constraint 3.16 directly in the **MILP** would introduce a quadratic constraint, resulting in a Mixed-Integer Quadratically Constrained Program (MIQCP) with substantially higher computational cost. To avoid this, we enforce a constant value for $\dot{h}$ during interactions and apply the constraint only to $\dot{r}$, sacrificing a small amount of flexibility for a significant gain in efficiency.

feasible trajectories with bounded accelerations for both robots and objects is a key design consideration. However, due to the chosen trajectory parametrization, directly constraining accelerations is not possible. The trajectory is defined by two sets of curves, and taking the second derivative of equation 2.26 results in non-convex expressions, rendering them unsuitable for inclusion in a **MILP** formulation.

To address this limitation, accelerations are instead attenuated in the cost function, following the approach in [10, 22], by introducing a quadratic penalty term

$$J(r_{S_i}, h_{S_i}) = \sum_{k=0}^{N-1} \ddot{r}_{S_i}(k)^\top Q_{\ddot{r}} \ddot{r}_{S_i}(k) + \ddot{h}_{S_i}(k)^\top Q_{\ddot{h}} \ddot{h}_{S_i}(k), \quad (3.17)$$

which is added to the cost function in equation 3.8.

In addition to the terms discussed above, the offline planning problem incorporates several further constraints to ensure feasibility, safety, and task compliance. These include constraints related to interaction timing and collision avoidance. For clarity of presentation, these additional components are omitted from the main formulation. A comprehensive overview of all constraints and objectives used in the offline planner is provided in Appendix X.

3.3 Online re-planner

Following the methodology in [10], an online planner is introduced to address limitations inherent in the offline plan generation. The offline trajectories do not incorporate updated state information of the robots and objects during execution, nor do they account for their physical dimensions. Several modelling assumptions and simplifications are therefore required to keep the online problem computationally tractable; relaxing these assumptions would require solving a full **MIP** online, which is not sufficiently fast for reliable real-time operation.

As a result, the online re-planner is deliberately limited in scope. It neither re-evaluates global **STL** satisfaction nor reconstructs the obstacle and collision-avoidance structure encoded in the offline solution. Instead, its role is to locally adjust the *preceding* and *interaction* Bézier curves of a robot when an interaction with an object becomes imminent. These adjustments account for updated robot and object states, their physical dimensions, and ensure that the robot is positioned such that the resulting normal force transfer induces the

desired change in object velocity.

Replanning is triggered prior to an upcoming interaction between a robot and an object. Let the curve on which this interaction occurs be indexed by I . Note that only the robot's preceding and interaction curves are replanned, no object curves are modified. For the online planner, define the start and end times of an interaction as t_S and t_F , respectively. These correspond to the first and last temporal parameter values of the interaction curve I . Define the start of the re-planned *preceding* curve as follows:

$$\begin{aligned}\hat{h}_R^{(I-1)(0)} &= t \\ \hat{r}_R^{(I-1)(0)} &= P_R^t \\ \hat{v}_R^{(I-1)(0)} &= \dot{P}_R^t\end{aligned}\tag{3.18}$$

Where t denotes the time at which the robot state was measured. We then define the re-planned start and end times of the interaction curve

$$\begin{aligned}\hat{h}_R^{(I)(0)} &= \hat{t}_S \\ \hat{h}_R^{(I)(n)} &= \hat{t}_F\end{aligned}\quad t < \hat{t}_S\tag{3.19}$$

Define t_m as the time at which the object's state is measured. Since the object cannot move under its own volition, we assume it maintains a constant velocity. The predicted object state $\tilde{P}_O$ at any candidate interaction time $\hat{t}_S$ is therefore obtained by linear propagation:

$$\tilde{P}_O^{\hat{t}_S} = P_O^{t_m} + \dot{P}_O^{t_m} \cdot (\hat{t}_S - t_m)\tag{3.20}$$

Define the unit push direction u as follows. Note here that $v_O^{(I)(n)}$ comes from the offline planner and represents the desired object velocity at the end of the interaction:

$$\begin{aligned}\Delta v &= v_O^{(I)(n)} - \dot{P}_O^{t_m}, \\ u &= \frac{\Delta v}{\|\Delta v\|}.\end{aligned}\tag{3.21}$$

Since only positive normal forces can be transferred between the robot and the object, it is essential that the robot be positioned such that the applied normal force aligns with the desired unit push direction u . To ensure this, we impose the following constraints at the beginning of the interaction. First, the robot must be positioned at a distance of $\rho_R + \rho_O$ from the object along the direction u . Second, the robot's velocity must match the object's velocity at the start of

the interaction.

$$\begin{aligned}\hat{r}_R^{(I)(0)} &= \tilde{P}_O^{\hat{t}_S} - (\rho_r + \rho_O) \cdot u \\ \hat{v}_R^{(I)(0)} &= \dot{P}_O^{t_m}\end{aligned}\tag{3.22}$$

To ensure that the direction of the force applied to the object remains parallel to the unit push direction throughout the interaction, we enforce:

$$\begin{aligned}\Delta \hat{v}^i &= \hat{v}_R^{(I)(i+1)} - \hat{v}_R^{(I)(i)} \\ \Delta \hat{v}^i \times \Delta v &= 0\end{aligned}\quad \forall i \in [0, n-1]\tag{3.23}$$

Although no replanning is done for the object curves, we define the position and velocity of the object at the end of the interaction as follows:

$$\begin{aligned}\hat{r}_O^{(I)(n)} &= \hat{r}_R^{(I)(n)} + (\rho_R + \rho_O) \cdot u \\ \hat{v}_O^{(I)(n)} &= \hat{v}_R^{(I)(n)}\end{aligned}\tag{3.24}$$

We must also ensure that the object's re-planned trajectory remains parallel to the offline trajectory. Let $r_O^{(I_{next})(0)}$ denote the MILP-planned object position at the start of the next interaction, and let $r_O^{(I)(n)}$ and $v_O^{(I)(n)}$ denote the MILP-planned position and velocity at the end of the current interaction.

To guarantee that the object's trajectory at the end of the interaction is parallel with the offline plan, we impose the following conditions:

$$(r_O^{(I_{next})(0)} - r_O^{(I)(n)}) \times (r_O^{(I_{next})(0)} - \hat{r}_O^{(I)(n)}) = 0,\tag{3.25}$$

$$\hat{v}_O^{(I)(n)} \times v_O^{(I)(n)} = 0\tag{3.26}$$

Finally, to ensure that the re-planned trajectory remains as close as possible to the original offline plan, we introduce soft penalties on the deviations in t_S , t_F , r_O , and v_O . Incorporating these terms, we formulate the following **Quadratic Programming (QP)**:

$$\begin{aligned}\delta_{t_1} &= \|\hat{t}_S - t_S\|_2^2, \\ \delta_{t_2} &= \|\hat{t}_F - t_F\|_2^2, \\ \delta_p &= \|\hat{r}_O^{(I)(n)} - r_O^{(I)(n)}\|_2^2, \\ \delta_v &= \|\hat{v}_O^{(I)(n)} - v_O^{(I)(n)}\|_2^2,\end{aligned}$$

$$\begin{aligned}
& \min_{r_R, h_R} \quad \ddot{r}_R(s)^\top Q_{\ddot{r}} \ddot{r}_R(s) + \ddot{h}_R(s)^\top Q_{\ddot{h}} \ddot{h}_R(s) + \delta_{t_1} + \delta_{t_2} + \delta_p + \delta_v, \\
& \text{s.t.} \\
& \text{Eqs. (3.3), (3.12), (3.16), (3.18), (3.19), (3.22)-(3.26).}
\end{aligned} \tag{3.27}$$

Note that the **QP** in (3.27) is only applied to interactions for which the desired object velocity at the end of the interaction is non-zero, i.e. $v_O^{(I)(n)} \neq \mathbf{0}$. If, on the other hand, $v_O^{(I)(n)} = \mathbf{0}$, then constraints (3.25) and (3.26) are replaced by

$$\hat{v}_O^{(I)(n)} = \mathbf{0}.$$

3.4 MPC

3.4.1 Approach

The purpose of the **MPC** is to enable real-time control of the robots, allowing the transportation problem to be executed on a physical free-flyer platform. Since the overall task can be divided into two distinct phases, interaction and non-interaction, we introduce two **MPC**: a nominal controller and a reactive controller. The nominal **MPC** governs trajectories without interactions and ensures accurate tracking of the planned motion. In contrast, the reactive **MPC** explicitly models robot–object interactions and provides low-level feedback to ensure that the object follows its reference trajectory during contact.

3.4.2 Force and Rate Model of a Free-Flyer

In this work, the robots are represented as rigid bodies operating in the 2D plane and actuated through forces and rate inputs, consistent with the formulation in [10]. Let m denote the robot mass. The free-flyer dynamics are given by

$$\dot{p} = v, \tag{3.28}$$

$$\dot{v} = \frac{1}{m} R(q)^T F, \tag{3.29}$$

$$\dot{q} = \frac{1}{2} \Xi(q) \tau, \tag{3.30}$$

where the robot state is $x_R = [p, v, q]$ and the input vector is $u = [F^T, \tau]$. Here, $F \in \mathbb{R}^2$ is the applied force in the body frame, and $\tau \in \mathbb{R}$ is the torque input. The matrix $R(q)$ denotes the rotation from the body frame to the world frame, while $\Xi(q)$ maps angular velocity to the corresponding quaternion rate.

This modeling approach captures linear motion and orientation but does not include rotational velocity states; instead, orientation evolves directly through the torque-driven quaternion rate equation.

3.4.3 Nominal MPC

To ensure that the robots accurately track their reference trajectories (Bezier curves), we define the nominal MPC problem as follows:

$$J^*(\tilde{x}(k)) = \min_{\mathbf{u}_k} J(x(n|k), u(n|k)), \quad (3.31)$$

$$\text{s.t. } x(m+1|k) = g(x(m|k), u(m|k)), \quad (3.32)$$

$$x(m|k) \in \mathbb{X}, \quad (3.33)$$

$$u(m|k) \in \mathbb{U}, \quad \forall m \in \mathbb{N}_{[0, N-1]}, \quad (3.34)$$

$$x(0|k) = x(k), \quad \forall n \in \mathbb{N}_{[0, N]}, \quad (3.35)$$

where $g(x(m|k), u(m|k)) = \text{RK4}[(3.28) - (3.30)]$ represents the discrete-time dynamics of the system. Since the online planner does not explicitly handle collision avoidance, we enforce safety between robots and objects during non-interacting trajectories using a **Control Barrier Function (CBF)**.

The cost function in equation (3.31) is described in detail in the Appendix, but in essence, for each shooting stage $m \in \mathbb{N}_{[0, N]}$, we sample the robot's Bezier curve to construct a reference trajectory

$$\mathbf{x}_R^{\text{ref}} = [\mathbf{p}_R^{\text{ref}}, \mathbf{v}_R^{\text{ref}}, \mathbf{q}_R^{\text{ref}}] \quad (3.36)$$

which is then used in the cost function. Since neither the **MILP** planner nor the online planner considers orientation explicitly, we define q^{ref} as the zero quaternion at all stages.

3.4.4 Reactive MPC

To capture the interaction between the robot and the object, we model the robot as directly influencing the object's linear accelerations. Angular acceleration and orientation of the object are not included, as their incorporation would

considerably complicate the formulation. Although the MPC does not operate directly on Bézier curves but instead on sampled points along these curves, we retain the established notation. For example, $h_O^{(I)(i)}$ denotes a sampled point rather than an actual control point. Note that $v_O^{(I)(n)}$ denotes the desired velocity of the object at the end of the interaction. We begin by constructing the object's reference trajectory from the robot's reference trajectory as follows:

$$\begin{aligned}
 \Delta v &= v_O^{(I)(n)} - \dot{P}_m^t \\
 u &= \frac{\Delta v}{\|\Delta v\|} \\
 h_O^{(I)(i)} &= h_R^{(I)(i)} \quad \forall i \\
 r_O^{(I)(i)} &= r_R^{(I)(i)} + u \cdot (\rho_R + \rho_O) \\
 v_O^{(I)(i)} &= v_R^{(I)(i)}
 \end{aligned} \tag{3.37}$$

The state vector of the reactive MPC consists of the concatenated robot state $x_R = [p_R, v_R, q_R]$ and object state $x_O = [p_O, v_O]$. Using the constructed trajectory in 3.37, we define the sampled reference trajectory of the object as

$$\mathbf{x}_O^{\text{ref}} = [\mathbf{p}_O^{\text{ref}}, \mathbf{v}_O^{\text{ref}}] \tag{3.38}$$

This reference includes only position and velocity, since the object's orientation is not modeled.

The interaction dynamics are based on the state derivatives in equations 3.28–3.30. Equation 3.28 remains unchanged as it lacks inputs, and equation 3.30 is unchanged due to the omission of object orientation. Substituting $m = 1$ into equation 3.29 yields the robot-generated force in the world frame, denoted by $\vec{F}_R$. The resulting interaction model is

$$\vec{\xi} = \vec{p}_O - \vec{p}_R \quad (3.39)$$

$$\delta = \nu(\|\vec{\xi}\| - \rho_R - \rho_O - \gamma) \quad (3.40)$$

$$\eta = \frac{1 - \tanh(\delta/2)}{2} \quad (3.41)$$

$$\vec{n} = \frac{\vec{\xi}}{\|\vec{\xi}\|} \quad (3.42)$$

$$p_s = \eta(\vec{n} \cdot \vec{F}_R) \quad (3.43)$$

$$P_s = \frac{1}{2}(p_s + \sqrt{p_s^2}) \quad (3.44)$$

$$\vec{F}_T = \vec{n} P_s \quad (3.45)$$

$$\dot{v}_O = \vec{F}_T / m_O \quad (3.46)$$

$$\dot{v}_R = \frac{\vec{F}_R - \vec{F}_T}{m_R} \quad (3.47)$$

Here, $\nu = 100$ is the activation gain, $\gamma = 0.005$ is the proximity threshold, ρ_R and ρ_O are the robot and object radii, and m_R, m_O their masses. Equations 3.39–3.41 define the activation function η , which signals when the robot is sufficiently close to transfer force. Equation 3.43 projects $\vec{F}_R$ onto the normal $\vec{n}$ and is modulated by η , while equation 3.44 ensures only positive normal forces are applied. Equations 3.45–3.47 describe the transmitted force and the resulting accelerations of the object and robot. These interaction dynamics are then incorporated into the system model $g(x(m|k), u(m|k))$, enabling the formulation of the reactive MPC problem as given in equations 3.31–3.35. The complete cost function of the reactive MPC, defined in equation 3.31, is described in detail in Appendix X.

Chapter 4

Results and Analysis

To evaluate the proposed method, experiments are conducted in a simulated two-dimensional microgravity analog environment based on the ATMOS [26] platform at **KTH Royal Institute of Technology (KTH)**. This platform consists of robots supported by air bearings to emulate the microgravity conditions of space, and actuated by pneumatic thrusters that enable both linear and angular motion in two dimensions.

All simulations are performed using a high-fidelity simulator that accurately captures the physical behavior of the platform. One notable simplification is the assumption of frictionless interaction between the robots. This assumption is made primarily to simplify the modeling process, as accurately modeling the physics of the air bearings would significantly increase the complexity of the simulation.

In addition to the simulation studies, the results are complemented by a hardware evaluation of the proposed method conducted on the ATMOS platform.

4.1 Simulations

A total of four transportation scenarios with varying levels of difficulty are simulated. For all scenarios, the nominal **MPC** is employed during non-interacting curves. To better highlight the effect of the reactive **MPC**, each scenario is executed for a total of 20 iterations. 10 iterations use the reactive **MPC** for interacting curves, while the remaining 10 iterations use the nominal **MPC** for both interacting and non-interacting curves.

For evaluation purposes, six performance metrics are considered, as summarized in Table 4.1. For Scenario 1, no obstacles are present; therefore,

Δv_{pre}	Velocity difference between robot and object at rendezvous
Δv_{post}	Object post-interaction velocity deviation
Δv_{Final}	Object velocity deviation at final time T
Δp_{Final}	Object position deviation at final time T
ΔT	Total task time deviation
$\Delta \rho$	Spatial robustness deviation

Table 4.1: Metrics

the metric ρ is omitted. All metrics, except for Δv_{pre} , are evaluated relative to the offline-generated **MILP** plan.

Providing a direct quantitative comparison for the online planner is not straightforward, as it actively modifies trajectories to account for the physical dimensions of the robots and to attenuate accelerations. Without this replanning step, the method would not be feasible, as the generated plans would continue to treat the robot and object as point masses. Instead, a qualitative evaluation of the online planner is presented through trajectory plots that illustrate how the planned motion is adapted to accommodate these physical constraints.

4.1.1 Basic Transportation

Shown in Figure 4.1(a) is the setup for scenario 1. The transportation task involves robots R_1 and R_2 transporting the object O_1 from its starting position to the goal. This scenario excludes any obstacles or **RoI**, as such no STL formulas are used. The total allotted time for the task is $T = 65$ seconds and the initial conditions are as follows: $R_1(0) = [2, 2]$, $R_2(0) = [20, 20]$, $O_1(0) = [4, 4]$. The goal of the task is $O_1(T) = [25, 25]$. An illustrative example of the original, re-planned and realized trajectories using the reactive **MPC** is shown in figures 4.1(b)-4.1(d).

The performance metrics for this scenario are presented in Table 4.2. The low values of Δv_{pre} observed for both **MPCs** indicate gentle contact between the robot and the object during the rendezvous phase. A comparison between the data collected using the reactive and nominal **MPC**, respectively, shows that the reactive **MPC** more effectively achieves the desired object velocity during interactions. Furthermore, the reactive **MPC** demonstrates improved accuracy in guiding the object to its desired goal position, indicating superior interaction handling and final positioning performance in this scenario.

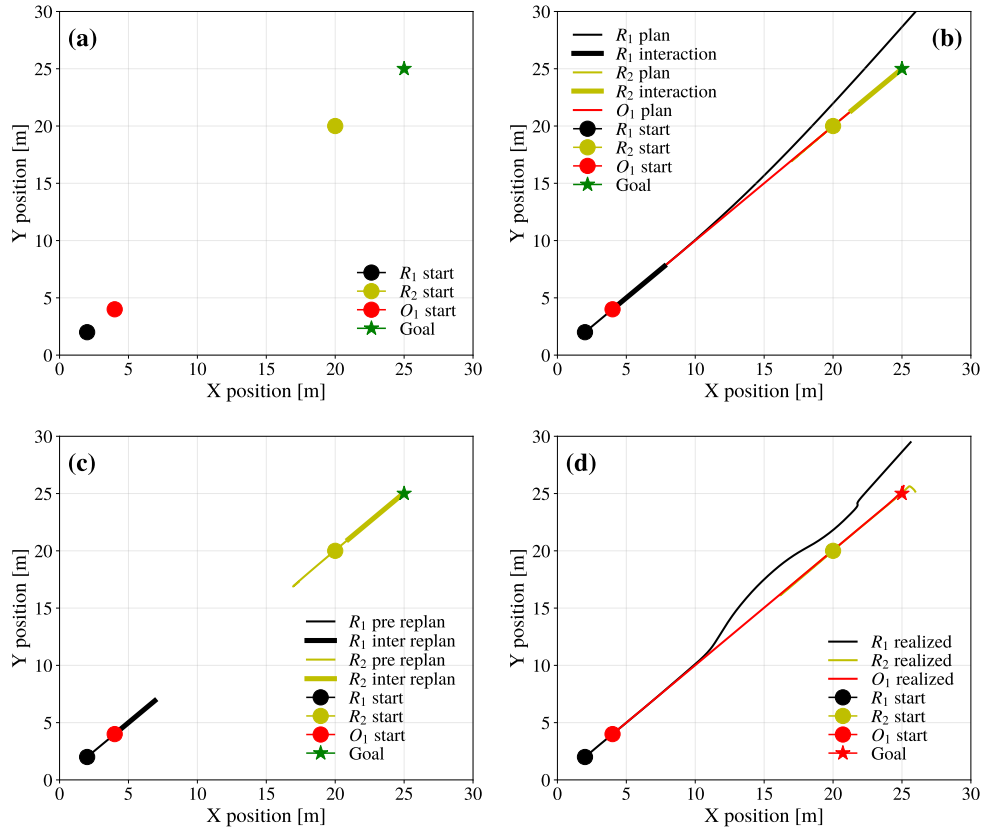


Figure 4.1: An example of scenario 1 with (a) the setup for the task. (b) the original plans produced by the MILP. (c) the re-planned trajectories of robots during task execution. (d) the final realized trajectories of all robots and objects.

4.1.2 Transportation & Obstacle Avoidance

To evaluate the performance of the proposed method on non-linear trajectories, two scenarios are designed and simulated, each containing an obstacle located between the initial and goal positions of the object. These scenarios are intended to assess both the method's ability to operate in environments with obstacles and hazards, as well as its performance when interactions induce changes in the object's direction rather than simple acceleration or deceleration along a straight-line path.

Both scenarios share similar initial conditions for the robots and the object, given by $R_1(0) = [2, 2]$, $R_2(0) = [20, 20]$, and $O_1(0) = [4, 4]$ for scenario 2 and $R_1(0) = [1, 4]$, $R_2(0) = [2, 9]$, and $O_1(0) = [3, 3]$ for scenario 3.

	Reactive MPC		Nominal MPC	
	Mean	Median	Mean	Median
Δv_{pre}	0.014 m/s	0.013 m/s	0.054 m/s	0.023 m/s
Δv_{post}	0.076 m/s	0.073 m/s	0.213 m/s	0.128 m/s
Δv_{Final}	0.078 m/s	0.066 m/s	0.307 m/s	0.227 m/s
Δp_{Final}	0.469 m	0.427 m	3.140 m	3.128 m
ΔT	10.230 s	10.336 s	11.300 s	12.135 s
$\Delta \rho$	-	-	-	-

Table 4.2: Metrics Scenario 1

The corresponding setups are shown in Figures 4.2(a) and 4.3(a), with task horizons of $T = 100$ and $T = 90$ for Scenarios 2 and 3, respectively. The objective in both tasks is for the object to reach the final goal position $O_1(T) = [25, 25]$, while ensuring obstacle avoidance for all systems throughout the execution:

$$\varphi = \bigwedge_{S_i \in \mathcal{S}} \mathbf{G}_{[0,T]}(p_{S_i} \in \mathcal{W}_{free})$$

Figures 4.2(b)–4.2(c) and 4.3(b)–4.3(c) present representative comparisons between the original plans and the re-planned trajectories for scenarios 2 and 3, respectively. In both cases, the results illustrate the effect of the online planner accounting for the physical size of the object as well as its updated position and velocity. This leads to a shift in the interaction locations while still aiming to minimize deviations of the object’s trajectory from the originally planned path.

Similar to Scenario 1, the metrics reported in Tables 4.3 and 4.4 indicate improved performance when using the reactive MPC. In both scenarios, the reactive MPC achieves smaller deviations in Δv_{post} and Δp_{Final} relative to the original plans compared to the nominal MPC.

The comparison of $\Delta \rho$, however, is less straightforward, as neither the reactive nor the nominal MPC exhibits a clear advantage. This behavior is expected, since neither the re-planner nor the MPC explicitly addresses obstacle avoidance. Instead, $\Delta \rho$ primarily serves as an indicator of the overall deviation of the executed trajectories from the original plans. While a small number of outliers lead to larger deviations in some cases, the results suggest that the difference in robustness between the planned and realized trajectories remains minor overall.

A closer inspection of Figures 4.2(d) and 4.3(d) reveals several perturba-

tions in the realized robot trajectories when compared to the original plans. Notably, Figure 4.3(d) shows a pronounced kink in the trajectory of R_1 . This behavior arises from the **CBF**, which enforce a minimum separation distance of 1 m between the robot and the object outside of interaction phases.

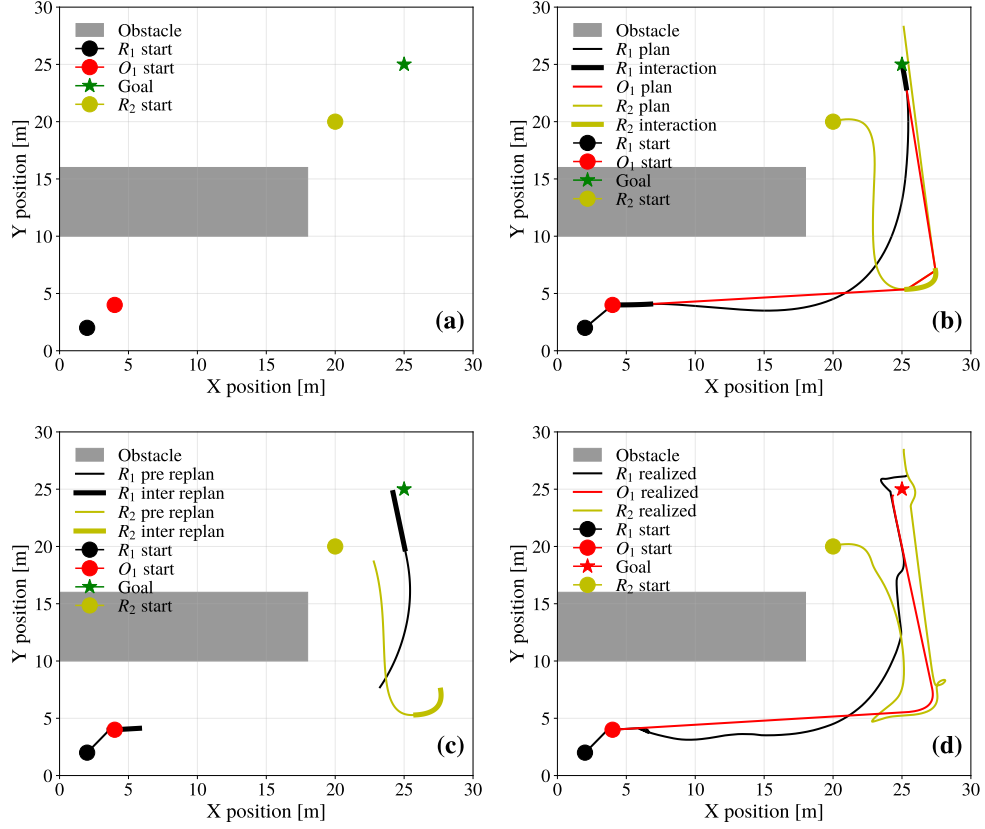


Figure 4.2: An example of scenario 2 with (a) the setup for the task. (b) the original plans produced by the **MILP**. (c) the re-planned trajectories of robots during task execution. (d) the final realized trajectories of all robots and objects.

4.1.3 Transportation with Waypoint Regions

The final scenario evaluates the performance of the proposed method in the presence of intermediary goals during transport, which also serves to assess its behavior under a more complex STL specification. Figure 4.4 illustrates the setup for scenario 4. The initial conditions are $R_1(0) = [2, 2]$, $R_2(0) = [1, 5]$, and $O_1(0) = [2, 4]$. The task is to transport O_1 to the goal position at $T = 122$,

	Reactive MPC		Nominal MPC	
	Mean	Median	Mean	Median
Δv_{pre}	0.019 m/s	0.017 m/s	0.053 m/s	0.023 m/s
Δv_{post}	0.105 m/s	0.063 m/s	0.152 m/s	0.202 m/s
Δv_{Final}	0.040 m/s	0.037 m/s	0.050 m/s	0.009 m/s
Δp_{Final}	0.814 m	0.822 m	6.116 m	6.297 m
ΔT	25.479 s	24.951 s	26.242 s	23.369 s
$\Delta \rho$	-0.001	-0.001	-0.120	-0.003

Table 4.3: Metrics Scenario 2

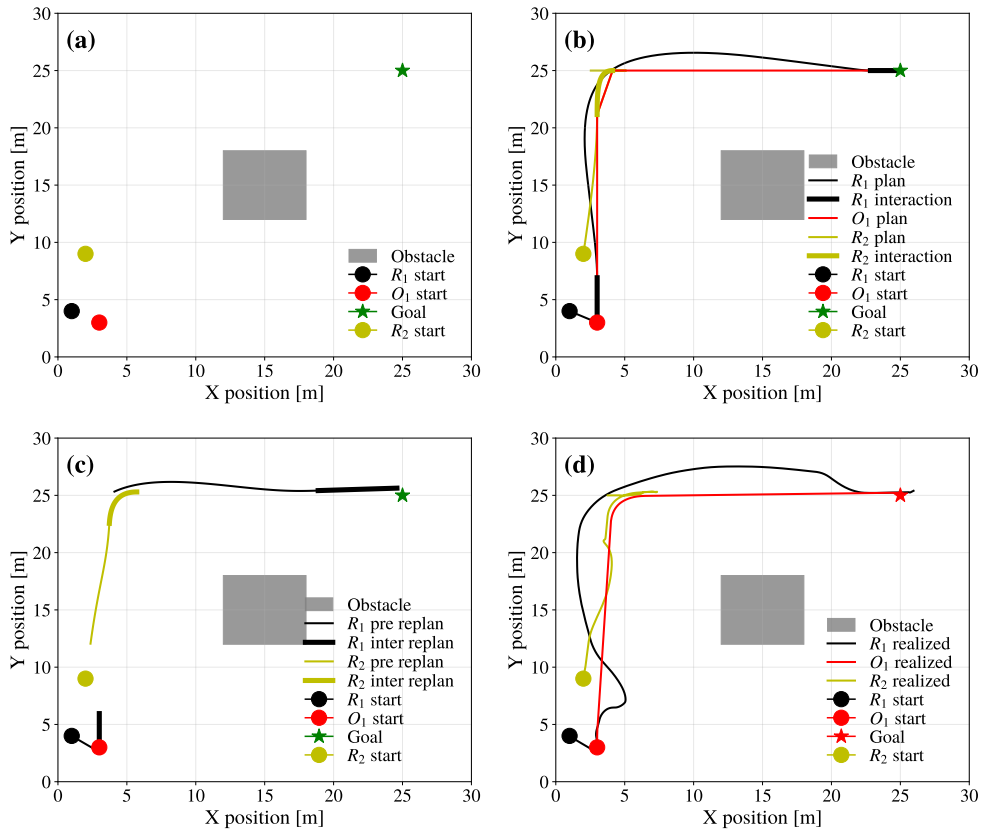


Figure 4.3: An example of scenario 3 with (a) the setup for the task. (b) the original plans produced by the MILP. (c) the re-planned trajectories of robots during task execution. (d) the final realized trajectories of all robots and objects.

$O_1(T) = [25, 25]$, while visiting two regions of interest, $RoI_1 = [10, 7]$ and $RoI_2 = [19, 21]$, each defined as a square of side length 2. The specification

	Reactive MPC		Nominal MPC	
	Mean	Median	Mean	Median
Δv_{pre}	0.030 m/s	0.022 m/s	0.061 m/s	0.038 m/s
Δv_{post}	0.260 m/s	0.073 m/s	0.329 m/s	0.211 m/s
Δv_{Final}	0.058 m/s	0.056 m/s	0.024 m/s	0.016 m/s
Δp_{Final}	0.670 m	0.515 m	4.412 m	3.332 m
ΔT	32.173 s	32.925 s	44.996 s	46.784 s
$\Delta \rho$	-0.523	+0.009	-0.108	-0.094

Table 4.4: Metrics Scenario 3

is given by

$$\varphi = \mathbf{F}_{[0,60]}(p_{O_1} \in RoI_1) \wedge \mathbf{F}_{[80,T]}(p_{O_1} \in RoI_2)$$

The metrics shown in Table 4.5 indicate comparable performance between the reactive and nominal MPC, with a slight robustness advantage observed for the nominal controller. In contrast to the other scenarios, no metric demonstrates a consistent advantage of the reactive MPC over the nominal formulation.

	Reactive MPC		Nominal MPC	
	Mean	Median	Mean	Median
Δv_{pre}	0.023 m/s	0.016 m/s	0.031 m/s	0.020 m/s
Δv_{post}	0.026 m/s	0.026 m/s	0.047 m/s	0.027 m/s
Δv_{Final}	0.037 m/s	0.043 m/s	0.082 m/s	0.012 m/s
Δp_{Final}	0.552 m	0.560 m	1.108 m	0.500 m
ΔT	5.481 s	5.170 s	4.693 s	3.711 s
$\Delta \rho$	-0.107	-0.110	-0.082	-0.037

Table 4.5: Metrics Scenario 4

4.2 Analysis

4.2.1 High-Level Performance Trends

An overview of all performance metrics across the evaluated scenarios reveals a consistent advantage of the reactive MPC over the nominal formulation. In particular, the reactive controller achieves lower mean values and reduced

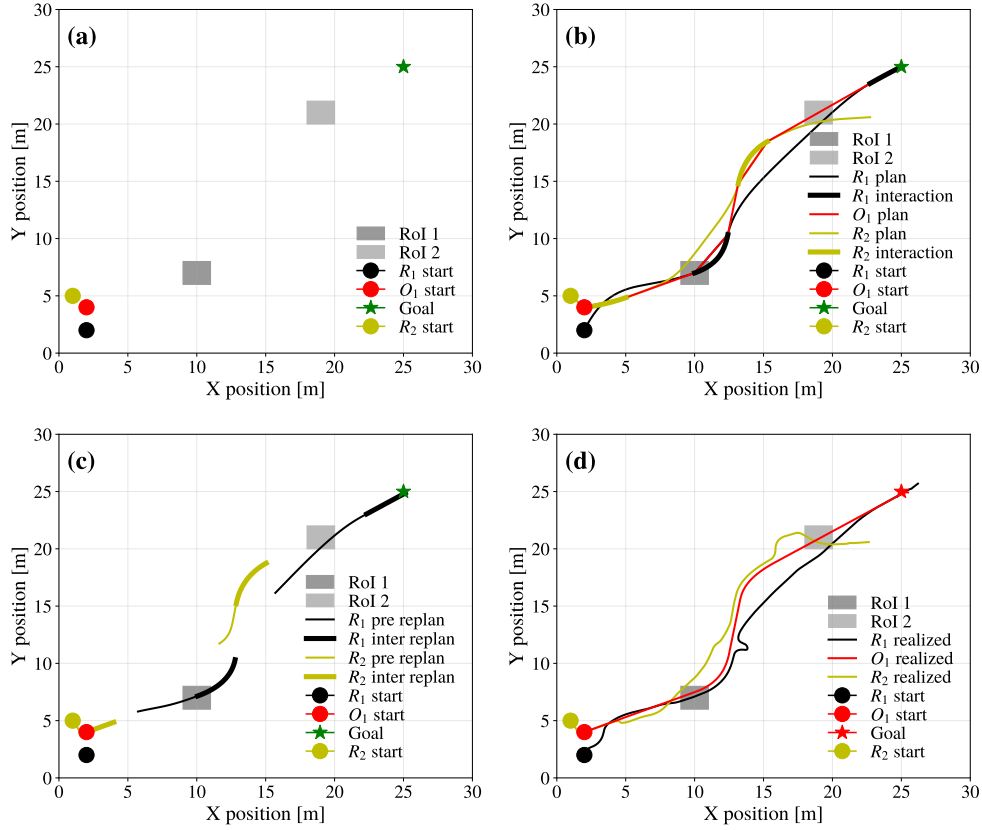


Figure 4.4: An example of scenario 4 with (a) the setup for the task. (b) the original plans produced by the **MILP**. (c) the re-planned trajectories of robots during task execution. (d) the final realized trajectories of all robots and objects.

variance for the interaction-related metrics Δv_{pre} , Δv_{post} , and the task-level metric Δp_{Final} . These improvements are most pronounced in scenarios that require higher object accelerations (Scenario 1) or induce directional changes in the object motion (Scenarios 2 and 3). In contrast, scenario 4, characterized by low interaction forces and gradual object motion, exhibit smaller performance differences between the two controllers. Beyond lower mean errors, the reduced variance observed with the reactive **MPC** indicates more consistent interaction outcomes across executions, which is critical for reliable autonomous operation.

4.2.2 Interaction Quality and Momentum Transfer

4.2.2.1 Rendezvous

A key indicator of interaction quality is the velocity difference between the robot and the object at rendezvous, quantified by Δv_{pre} . Across all scenarios, the reactive **MPC** consistently achieves both lower mean values and reduced variance of this metric compared to the nominal controller. This indicates gentler and more reliable initial contact during interaction phases. The improved performance can be attributed to the reactive controller's explicit consideration of object dynamics, which enables better velocity matching prior to contact. In contrast, the nominal **MPC** does not model the object dynamics, resulting in less accurate anticipation of the required interaction conditions.

4.2.2.2 Post-interaction Velocity

The advantage of the reactive **MPC** becomes even more pronounced when examining the post-interaction velocity deviation Δv_{post} . Across all scenarios, the reactive controller exhibits a substantial reduction in both the mean and variance of this metric relative to the nominal formulation. This suggests that the reactive controller more accurately anticipates the momentum transfer during interaction and is therefore better able to regulate the object's post-interaction trajectory. The effect is particularly evident in scenarios requiring directional changes of the object (Scenarios 2 and 3), where the nominal controller struggles to handle multi-dimensional accelerations arising from unmodeled object dynamics. In contrast, the reactive controller is better suited to capture these effects, resulting in post-interaction trajectories that more closely follow the planned behavior.

4.2.3 Final Object State Accuracy

4.2.3.1 Position Accuracy

The final position deviation Δp_{Final} aggregates the cumulative positional error of the object relative to the original plan and constitutes the most critical task-level metric, as successful mission completion ultimately depends on delivering the object to its desired goal location. In Scenarios 1–3, the reactive **MPC** achieves an order-of-magnitude improvement in this metric compared to the nominal controller, with the largest gains observed in scenarios involving directional changes.

These results indicate that errors introduced during interaction phases dominate the final positional error of the object. Once such errors occur, neither the online re-planner nor the **MPC** can fully compensate for them retroactively. For instance, if an interaction imparts an incorrect post-interaction velocity shortly before task completion, the controller can at best decelerate the object along its resulting trajectory, but cannot redirect it sufficiently to ensure convergence to the desired goal within a tight tolerance. Consequently, limiting errors during interaction phases is essential for achieving accurate final positioning. The reactive controller's improved velocity matching and post-interaction regulation, therefore, plays a critical role in reducing error accumulation over successive interactions.

4.2.3.2 Velocity Accuracy at Task Completion

The final velocity deviation Δv_{Final} exhibits a less uniform trend across scenarios. In several cases, particularly Scenarios 2–4, the nominal **MPC** achieves comparable or lower final velocity errors than the reactive formulation. This behavior can be explained by the structure of the tasks, in which the desired final velocity of the object is zero. As a result, the final interaction always involves decelerating the object to rest.

While the reactive controller provides increased flexibility during interaction phases, this flexibility is most beneficial when the object initially has low momentum or when directional changes are required. In scenarios dominated by deceleration, the nominal controller can perform competitively, provided that the imparted object velocity prior to the final interaction is not excessively large. This effect is clearly illustrated in Scenario 1, where the nominal **MPC** imparts a significantly higher object velocity due to unmodeled dynamics, resulting in a larger deceleration demand during the final interaction and consequently a larger Δv_{Final} . In contrast, the reactive controller limits the imparted velocity earlier in the task, reducing the burden on the final interaction.

4.2.4 Task Duration

Across all scenarios and both controllers, a significant deviation in task duration ΔT is observed, in some cases exceeding 50% of the nominal task horizon. This behavior is primarily attributed to the absence of explicit acceleration constraints in both the offline planning and online control stages. As discussed in Section 3.2.3.5, direct modeling and constraint enforcement

on accelerations is not feasible within the adopted framework. Instead, accelerations are indirectly attenuated through cost terms.

To ensure smooth and stable interactions, accelerations are further penalized in the online re-planner, resulting in lower object velocities and consequently longer execution times. This accounts for a substantial portion of the increased ΔT observed for the reactive controller, as well as part of the reduction in Δv_{post} . A similar effect is present for the nominal controller; however, its inability to model object dynamics leads to larger post-interaction velocity deviations, which in turn require additional corrective actions and further prolong task execution. While the resulting increase in task duration represents a limitation of the approach, it reflects a deliberate trade-off between temporal efficiency and the robustness and feasibility of physical interactions.

4.2.5 Spatial Robustness

As previously noted, the spatial robustness metric $\Delta\rho$ does not directly measure the **STL** satisfaction performance of the online planner or the **MPC**, as neither explicitly enforces STL constraints. Instead, $\Delta\rho$ captures the deviation of the realized trajectories from the robustness margins of the original offline-generated plans. The generally low mean values and variance of $\Delta\rho$ across scenarios and controllers indicate that the proposed method preserves STL satisfaction margins reasonably well. The observed outliers are primarily attributable to the activation of **CBFs** within the **MPC**, which can introduce localized deviations from the original trajectories while maintaining safety constraints.

4.2.6 Regime-Dependent Effectiveness

In contrast to Scenarios 1–3, Scenario 4 exhibits comparable performance between the reactive and nominal controllers across most metrics. Although no definitive conclusion can be drawn, a plausible explanation is that tasks characterized by low interaction forces and small object accelerations can be handled effectively by the nominal **MPC**. In this operating regime, the mismatch between required and modeled interaction forces remains small, as the object experiences only mild accelerations. Consequently, the robot does not operate near actuator saturation, allowing the nominal controller to compensate for the relatively minor unmodeled dynamics.

These results suggest that the benefits of the reactive controller scale with interaction intensity and object acceleration. For tasks involving

minimal acceleration and gentle interactions, the nominal controller may be sufficient or even preferable due to its lower complexity, whereas the reactive formulation provides clear advantages in more dynamically demanding regimes.

4.3 Experiments/Hardware Validation

Chapter 5

Discussion

Mention that because of the complex nature of the MILP, adding STL constraints caused the complexity to grow too much, making it unpractical to solve since the solve time became extremely long. This is the reason that most of the constraints, like the goal position were not encoded using STL and instead directly using MILP, only collision avoidance was encoded using STL, and it caused a significant increase in computation time. Joris paper showed how complex STL formulas can be used with MILP. In order for STL constraint to work better for this method, we could for instance rewrite STL constraints to use fewer binary variables as was done in [27]. For this same reason, we did not encode temporal robustness and instead provided the metric delta t final to illuminate how the timing of the plan was shifted.

Mention for the MPC that using just min,max for the activation function results in no gradients which doesn't work well.

Were assuming no friction in the simulations which is a big assumption, and does not model real world well. To better validate it should be extended to not assume zero friction in the simulation, and also make the MPC model friction, orientation and angular velocity of the object.

One major thing is that the interactions are very unstable, as is expected for such an underactuated coupling. Especially because both objects and robots are assumed circular, this could be compensated by for instance ... see below.

One way to counteract the inherent unstableness of the interaction (like balancing on a knife's edge because of circular geometries) would be to introduce some compliant material around the robots, or equip them with a cone or half circle made for engaging the objects and allowing more than a single contact point. This would reduce the unstableness and could also

mitigate the transfer of angular accelerations.

Discuss why the re planner was constructed the way it was. because a more explicit encoding would cause a significant amount of non linearities to be part of program, making the use of QP solvers impossible and thus having to resort to non-linear solvers which in general are much slower. For instance, instead of all the bending over backwards to ensure parallelism with the offline plan one could simply enforce that the end velocity and position of the object at the end of an interaction would be such that it reaches the next interaction position after some time. But this would in turn make the ΔV variable and subsequently the unit push direction u into decision variables, turning the QP into a non linear

DONT FORGET A BIG THING; ALL OF THE MEASUREMENTS WERE FROM ground truth and not sensors that have noise and shit. We should need to add a perception system to estimate the pose of objects and other robots and see how well this performs.

Chapter 6

Conclusions and Future work

6.1 Conclusions

6.2 Limitations

6.3 Future work

3D modeling, orientation and angular velocity of the object. Model friction, improve planners. Make non linearity work. Improve the MPC, better STL formulations.

6.4 Reflections

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If you do not have an appendix, do not include the \cleardoublepage command below; otherwise, the last page number in the metadata will be one too large.

Appendix A

Supporting materials

Here is a place to add supporting material that can help others build upon your work. You can include files as attachments to the PDF file or indirectly via URLs. Alternatively, consider adding supporting material uploaded as separate files in DiVA.

The BibTeX references used in this thesis are attached. 

Some source code relevant to this project can be found at <https://github.com/ggmaquirejr/E-learning> and <https://github.com/ggmaquirejr/Canvas-tools>.

Your reader can access the attached (embedded) files using a PDF tool such as Adobe Acrobat Reader using the paperclip icon in the left menu, as shown in Figure A.1 or by right-clicking on the push-pin icon in the PDF file and then using the menu to save the embedded file as shown in Figure A.2.

An argument for including supporting material in the PDF file is that it will be available to anyone who has a copy of the PDF file. As a result, they do not have to look elsewhere for this material. This comes at the cost of a larger PDF file. However, the embedded files are encoded into a compressed stream within the PDF file; thus, reducing the number of additional bytes. For example, the references.bib file that was used in this example is 10 617 B in size but only occupies 4 261 B in the PDF file.

DiVA is limited to ≈ 1 GB for each supporting file. If you have very large amounts of supporting material, you will probably want to use one of the data repositories. For additional help with this, contact KTH Library via researchdata@kth.se.

As of Spring 2024, there are plans to migrate this supporting data from DiVA to a research data repository.

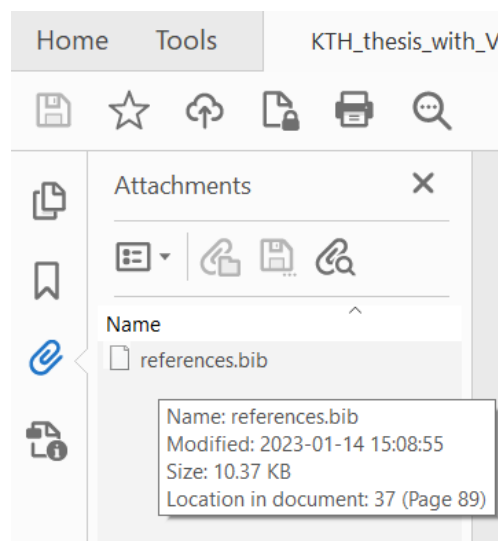


Figure A.1: Adobe Acrobat Reader using the paperclip icon for the attached references.bib file

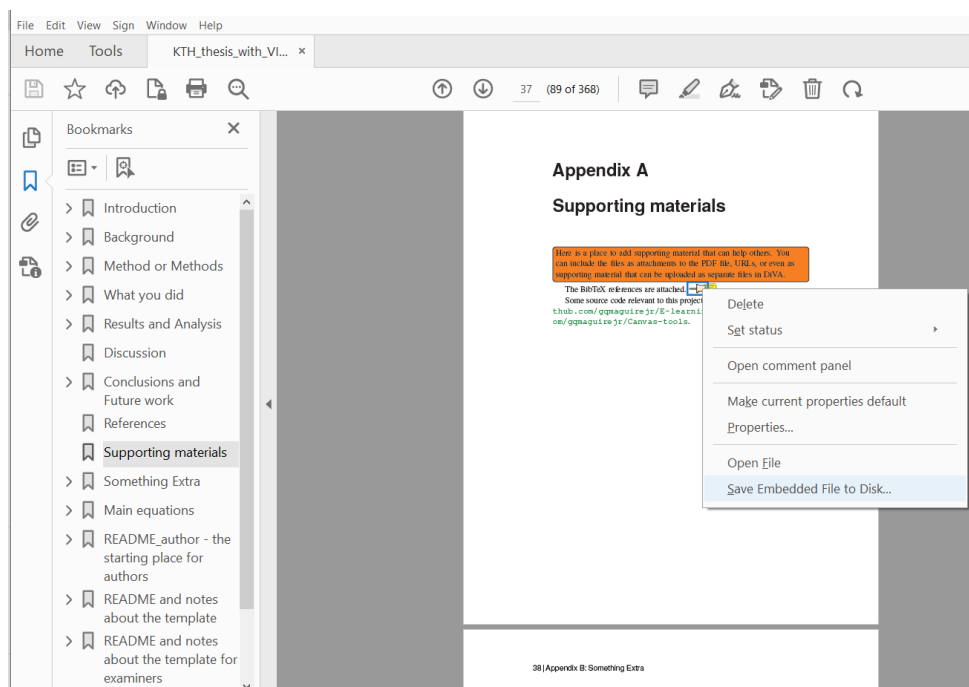


Figure A.2: Adobe Acrobat Reader after right-clicking on the push-pin icon for the attached references.bib file

Appendix B

Something Extra

svensk: Extra Material som Bilaga

B.1 Just for testing KTH colors

You have selected to optimize for print output

- Primary color

- kth-blue 

- kth-blue80 

- Secondary colors

- kth-lightblue 

- kth-lightred 

- kth-lightred80 

- kth-lightgreen 

- kth-coolgray 

- kth-coolgray80 

black 

Appendix C

README_author - the starting place for authors

This chapter describes the thesis template that I (Gerald Q. Maguire Jr.) have developed for use at KTH Royal Institute of Technology (KTH). It is important to note that the template is **not prescriptive**, as not every thesis will have all of the parts that the template shows. However, if there is something that you decide to leave out, you should make a conscious decision to do so and you should consider the impact this may have on your thesis being approved by the examiner.

Fundamental to the design of the template are several key factors:

- Helping students be successful in their degree project,
- Helping students produce a high-quality thesis, and
- Supporting all of the (relevant) phases of the degree project process.

This document is a work in progress.

C.1 Advice for Author or Authors

One of the hardest problems an author faces is getting started writing, *i.e.*, the blank sheet of paper – empty file barrier. The template provides a non-blank starting point; hence, avoiding the blank paper barrier. Additionally, the template provides some initial structure, basically, an Introduction, Methods, Results, and Discussion (IMRAD) structure, so that there are hints of where to place material. Moreover, there are places (and notes) about material that the

student should consider adding; for example, the “required reflections” section in the final chapter.

The template (located in the file `examplethesis.tex`) also provides some examples of commonly occurring types of content, so that one can easily find examples of how to include a figure, table, code listing, *etc.* These examples are not meant to be exhaustive and quite often the student will probably need to learn new $\text{\LaTeX}$ commands in the course of writing their thesis.

As an author, the first step is to configure the $\text{\LaTeX}$ engine that you will use to process the files - see Appendix C.2. The second step will be to configure the template - see Appendix C.3. The third step will be to make sure that the information about you, your supervisor(s), and the examiner are correct in the file `custom_configuration.tex` - this information uses the macros described in Appendix C.4. Now that you have a lot of the administrative details taken care of it is time to start to write - see Appendix C.6.

Note that if you are using Overleaf:

Make your own copy of the template If you have opened the template from a URL, in the upper left-hand corner, click on **Menu**. Then select **Copy Project** - this will give you your own private copy.

Use a helpful project name I suggest you include your name in the project name so that when you share it with your supervisor(s) and examiner, they will know it is your project.

Invite your supervisor(s) and examiner to your project You can invite your supervisor(s) and examiner to your project and they can directly comment on and correct your drafts.

Log in to Overleaf with your KTH account If you log in to Overleaf with your KTH account, you get a version of Overleaf that lets you turn on “Track changes” which is very useful (particularly if you have invited your supervisor(s) and examiner to join your project). It also gives you a bit more of a time budget to compile (which can be useful if you have a lot of Tikz figures or other things that take a lot of time for the $\text{\LaTeX}$ engine to render).

If you have more detailed questions about the template itself - see Appendix D.

C.2 Author configuration of the $\text{\LaTeX}$ engine

The template should work with $\text{\pdfLaTeX}$, $\text{\XeLaTeX}$, and $\text{\LuaLaTeX}$. If you are using Overleaf, I strongly recommend using $\text{\XeLaTeX}$ — as this will get the `Arial` fonts correct for the KTH cover. If you are running the compiler on your local machine and you use $\text{\XeLaTeX}$ **and** you have `Arial` as a system font, then it will be able to use it. Similarly, for $\text{\LuaLaTeX}$. For $\text{\pdfLaTeX}$ I have used `\fontfamilyhelvet`, *i.e.*, Helvetica, as it is a sans serif font.

One student reported problems with `FONTSPEC` not loading the fonts properly when running locally with macOS 12.4, TeXLive 2022, LaTeX Workshop on VS Code, and $\text{\XeLaTeX}$ - the solution is described at <https://tug.org/TUGboat/tb39-2/tb122robertson-fontspec.pdf>.

If you are using Overleaf, it is easy to select the compiler (*i.e.*, $\text{\TeX}$ engine) by using the drop-down menu, as shown in Figure C.1.

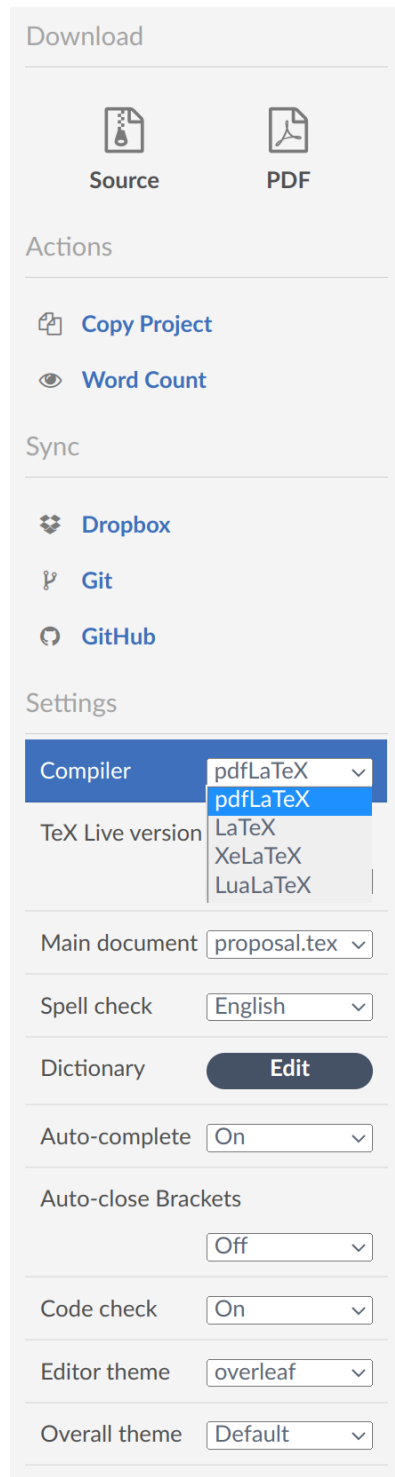


Figure C.1: Selecting a compiler (*i.e.*, TeX engine) in Overleaf

C.3 Author configuration of the template

The template is designed to handle a thesis written in English or Swedish. You can set the default language to ‘english’ or ‘swedish’ by passing an option to the documentclass. Note that the language option is written in all lowercase letters; for example, to set the document’s language to English:

```
\documentclass[english]{kththesis}
```

To set the document’s language to Swedish (uncomment the following line):

```
\documentclass[swedish]{kththesis}
```

The language option ‘swedish’ sets the conditional `\ifinswedish` to true. Among many other things, this conditional is used to configure the KTH cover and the title page to use the chosen language.

The two most common bibliographic engines are supported, *i.e.*, BibTeX and BibLaTeX. To set the language to English and use the bibliographic engine to BibTeX you would say:

```
\documentclass[english, bibtex]{kththesis}
```

To set the language to Swedish and use the bibliographic engine to BibLaTeX you would say:

```
\documentclass[swedish, biblatex]{kththesis}
```

The above illustrates that you can pass multiple options to the document class separated by commas. Also, note that the options were passed as all lowercase letters.

You can, of course, also modify the formatting of the citations and bibliography. See for example the following code snippet:

```
\ifbiblatex
% \usepackage[language=english, bibstyle=authoryear,
% ↪ citestyle=authoryear, maxbibnames=99]{biblatex}
% \usepackage[style=numeric, sorting=none, backend=biber
% ↪ ]{biblatex}
\usepackage[bibstyle=authoryear, citestyle=authoryear,
% ↪ maxbibnames=99, language=english]{biblatex}
% alternatively you might use another style, such as
% ↪ IEEE
% \usepackage[style=ieee]{biblatex}
\addbibresource{references.bib}
% \DeclareLanguageMapping{norsk}{norwegian}
```

```

\else
  % The line(s) below are for BibTeX
  \bibliographystyle{bibstyle/myIEEETran}
  %\bibliographystyle{apalike}
\fi

```

To optimize for digital output (this changes the color palette) add the option: `digitaloutput`. There are also options for A4 or G6 paper: `a4paper` or `g5paper` (respectively). There is an option for nomenclature, to produce and refer to equations. Finally, there are options for a 1st cycle thesis or 2nd cycle thesis: `bachelor` and `master` (respectively); however, these two options are **not** currently used.

One of the first things that the author(s) will want to do is add the working title and subtitle to the thesis. This is done using the `\title`, `\subtitle`, `\alttitle`, and `\altsubtitle` macros as shown below:

```

\title{This is the title in the language of the thesis}
\subtitle{A subtitle in the language of the thesis}

% give the alternative title - i.e., if the thesis is in
  ↪ English,
% then give a Swedish title
\alttitle{Detta är den svenska översättningen av titeln}
\altsubtitle{Detta är den svenska översättningen av
  ↪ undertiteln}
% alternative, if the thesis is in Swedish, then give an
  ↪ English title
%\alttitle{This is the English translation of the title}
%\altsubtitle{This is the English translation of the
  ↪ subtitle}

```

Setting these values once and then using them in many places reduces the work to change them while at the same time ensuring consistency.

The issue of what characters can be placed in a title (subtitle, keywords, and abstracts) and how to enter them is a bit complex, but you should be able to achieve much of what you want - at the cost of some effort. Ultimately, the characters that will be placed into DiVA and LADOK must be Unicode, encoded using Unicode Transformation Format (UTF), specifically UTF-8. While DiVA utilizes HTML for these fields, allowing you to include some markup, LADOK does not support any markup.

If you have titles or subtitles that cannot be converted to pure Unicode (such as subscripts with capital letters or other symbols), you should also use the optional commands `\titleInPlainText`, `\subtitleInPlainText`, `\alttitleInPlainText`, and `\altsubtitleInPlainText` in your `custom_configuration.tex` or `custom_configuration_plaintext.tex` files to provide versions of titles and subtitles that can be used with LADOK. Compiling `Generate_titles_in_plain_text.tex` with **Lua^AT_EX** will produce a file named `extra_plaintext.tex` containing plain text versions of the titles and subtitles (both main and alternate). The reason that you have to compile using Lua^AT_EX is that I have used Lua code and Lua^AT_EX to compute the plain text strings. After you have compiled the file, you can follow the workflow to generate, download, edit and merge, and upload the `custom_configuration_plaintext.tex` file to your Overleaf project.

Some additional configuration that the author(s) might do is to set the values of the macros related to the course cycle, course code, date of the thesis, number of credits, degree/exam name, subject area, and if the degree is done external to KTH to set the host information (see the file *custom_configuration.text*). Consider the snippet below for a student admitted to the “Bachelor’s Programme in Information and Communication Technology (TCOMK)” program and enrolled in the degree project course “IA150X Degree Project in Information and Communication Technology, First Cycle 15.0 credits” and working at a company “Företaget AB”:

```
\hostcompany{Företaget AB} % Remove this line if the
    ↪ project was not done at a host company

\date{\today}

\courseCycle{1}
\courseCode{IA150X}
\courseCredits{15.0}

\programcode{TCOMK}
\degreeName{Bachelors degree}
% Note that the subject area for a Bachelor's thesis (
    ↪ Kandidatexamen)
```

```
% should be either Technology or Architecture
% If the thesis is in Swedish, these would be: teknik |
    ↪ arkitektur
% -- Note the use of lower case for the Swedish subject
    ↪ area
\subjectArea{Technology}
```

Note that in the above macros you have to give the English or Swedish names in the arguments to `\degreeName` and `\subjectArea` - as shown below:

```
\degreeName{Kandidatexamen}
\subjectArea{teknik}
```

For a CDATE student enrolled in the course “DA231X Degree Project in Computer Science and Engineering, Second Cycle 30.0 credits”, the cycle, program, course code, degree, and subject area information would be:

```
\programcode{CDATE}
\courseCycle{2}
\courseCode{DA231X}
\courseCredits{30.0}
\degreeName{Degree of Master of Science in Engineering}
\subjectArea{Computer Science and Engineering}
```

The set of possible values for the English or Swedish names in the arguments to `\degreeName` are:

```
\degreeName{Higher Education Diploma}
\degreeName{Högskoleexamen}

\degreeName{Bachelors degree}
\degreeName{Kandidatexamen}

\degreeName{Master of Architecture}
\degreeName{Arkitektexamen}

\degreeName{Degree of Master of Science in Engineering}
\degreeName{Civilingenjörs}

\degreeName{Magister}
\degreeName{Magisterexamen}

\degreeName{Degree of Master of Science}
\degreeName{Masterexamen}
```

```

\degreeName{Master of Science in Engineering and Master
  ↳ of Arts in Education degree}
\degreeName{Civilingenjör och lärare examen}

\degreeName{Degree of Master of Science in Secondary
  ↳ Education}
\degreeName{Ämneslärarexamen}

\degreeName{Both} # Degree Project in the Field of
  ↳ Technology <teknikområde> and the Main Field of
  ↳ Study <huvudområde>
\degreeName{Same} # The case when the field of
  ↳ technology <teknikområde> and main field of study <
  ↳ huvudområde> are the same.

```

For the last two cases, the code compares the values of subjectArea and secondSubjectArea.

You can find a list of the program codes and school acronyms in the file: lib/schools_and_programs.ins.

There are a set of rules about what is to be displayed on the KTH cover. These can be found at <https://www.kth.se/social/group/sprakkommitte/page/omrade-for-examensarbete/>.

One of the reasons for many of the macros shown above and below is to collect the information that is needed to report the approved thesis in Digitala Vetenskapliga Arkivet (DiVA) and to report the title(s) and grade in Lokalt adb-baserat dokumentationssystem (LADOK).

National subject categories are a **required** field in the DiVA record. These categories follow a definition by SCB (nowadays known as Statistikmyndigheten or in English: Statistics Sweden) and HSV (Högskoleverket - nowadays known as Universitetskanslersämbetet (UK-ämbetet) and Universitets- och högskolerådet (UHR) or in English: Swedish Higher Education Authority and Swedish Council for Higher Education). While these codes refer to research areas, these codes are also used in KTH to indicate the area of the thesis. The guidance that I received from the Linköping University library was that one should try to use 5-digit codes when possible. Some examples of these codes are shown in Table C.1.

```

\nationalsubjectcategories{} comma-separated list of national
                             subject category codes - each a 3
                             or 5 digit code

```

An example for a thesis in Computer Science and Computer Systems:

```
\nationalsubjectcategories{10201, 10206}
```

You can find examples of the subjects and their codes at in Table C.1. The complete list can be found AT <https://www.scb.se/contentassets/3a12f556522d4bdc887c4838a37c7ec7/standard-for-svensk-indelning--av-forskningsamnen-2011-uppdaterad-aug-2016.pdf>

and

<https://www.scb.se/contentassets/10054f2ef27c437884e8cde0d38b9cc4/oversattningsnyckel-forskningsamnen.pdf>

You can also find them in README_notes/National_Subject_Categories.tex. This file can be processed separately or included as an appendix by commenting out the begin and end comments around it near the end of the examplethesis.tex file. The full list is about 14 pages long.

Table C.1: Examples of some national subject categories and their codes

Code	Category (in Swedish)	Category (in English)
102	Data- och informationsvetenskap (Datateknik)	Computer and Information Sciences
10201	Datavetenskap (datalogi)	Computer Sciences
10202	Systemvetenskap, informationssystem och informatik (samhällsvetenskaplig inriktning under 50804)	Information Systems (Social aspects to be 50804)
10203	Bioinformatik (beräkningsbiologi) (tillämpningar under 10610)	Bioinformatics (Computational Biology) (applications to be 10610)
10204	Människa-datorinteraktion (interaktionsdesign) (Samhällsvetenskapliga aspekter under 50803)	Human Computer Interaction (Social aspects to be 50803)
10205	Programvaruteknik	Software Engineering
10206	Datorteknik	Computer Engineering
10207	Datorseende och robotik (autonoma system)	Computer Vision and Robotics (Autonomous Systems)
10208	Språkteknologi (språkvetenskaplig databehandling)	Language Technology (Computational Linguistics)
10209	Medieteknik	Media and Communication Technology
10299	Annan data- och informationsvetenskap	Other Computer and Information Science
202	Elektroteknik och elektronik	Electrical Engineering, Electronic Engineering, Information Engineering
20201	Robotteknik och automation	Robotics
20202	Reglerteknik	Control Engineering
20203	Kommunikationssystem	Communication Systems
20204	Telekommunikation	Telecommunications
20205	Signalbehandling	Signal Processing
20206	Datorsystem	Computer Systems
20207	Inbäddad systemteknik	Embedded Systems
20299	Annan elektroteknik och elektronik	Other Electrical Engineering, Electronic Engineering, Information Engineering

C.3.1 Setting the UN's Sustainable Development Goals (SDGs) information

The UN’s Sustainable Development Goals (SDGs) are a field in the DiVA record. Some examples of these SDGs are shown in Table C.2.

\SDGs{ }	comma-separated list of SDG codes - each a numerical value
-----------------	---

An example of a thesis supporting SDGs 8 and 9:

$$\backslash \text{SDGs} \{8, 9\}$$

Table C.2: UN's Sustainable Development Goals (SDGs)

SDG	Category (in English)	Category (in Swedish)
1	No poverty	Ingen fattigdom
2	Zero hunger	Ingen hunger
3	Good health and well-being	God hälsa och välbefinnande
4	Quality education	God utbildning för alla
5	Gender equality	Jämställdhet
6	Clean water and sanitation	Rent vatten och sanitet för alla
7	Affordable and clean energy	Hållbar energi för alla
8	Decent work and economic growth	Anständiga arbetsvillkor och ekonomisk tillväxt
9	Industry, innovation and infrastructure	Hållbar industri, innovationer och infrastruktur
10	Reduced inequalities	Minskad ojämlikhet
11	Sustainable cities and communities	Hållbara städer och samhällen
12	Responsible consumption and production	Hållbar konsumtion och produktion
13	Climate action	Bekämpa klimatförändringarna
14	Life below water	Hav och marina resurser
15	Life on land	Ekosystem och biologisk mångfald
16	Peace, justice and strong institutions	Fredliga och inkluderande samhällen
17	Partnerships for the goals	Genomförande och globalt partnerskap

C.4 Author macros

It is assumed that there can only be 1 or 2 authors. For many years now 2nd cycle theses are expected to only have one author.

For the author or first author, there are a number of macros defined to store information about the author, so that it can later be used in multiple places – for example, the KTH cover (produced with `\kthcover`), the title page (via `\titlepage`, the “For DiVA” section at the end of the thesis (via `\divainfo{pg:lastPageofPreface}{pg:lastPageofMainmatter}`), and possibly a JavaScript Object Notation (JSON) file named `fordiva.json` produced as a by product of the `\divainfo`. Note that the actual section name has DiVA set in all caps - which hopefully should not occur in the thesis! If the string DiVA set in all caps, does have to appear, then the section heading should be preceded by four euro signs and followed by four more euro signs (as is done in this document).

The author-related macros are:

<code>\authorsLastname{ }</code>	the last name of the author*
<code>\authorsFirstname{ }</code>	the first name of the author
<code>\email{ }</code>	the KTH e-mail address of the author
<code>\kthid{ }</code>	the author’s kthid, this generally starts with the string “u1” and is a unique identifier for every KTH user.
<code>\authorsSchool{ }</code>	the value is generally of the form: <code>\schoolAcronym{XXX}</code> . The currently supported school acronyms are: ABE, CBH, EECS, ITM, and SCI. These are defined in the file <code>schools_and_programs.ins</code> . There is also a fake school XXX for use in the template.

If the first author is not in Stockholm, Sweden when the acknowledgements are written, then add that information via the macros described below. This information will be used when generating the acknowledgements

*Note that the author’s name can include a suffix such as “, Jr.” or “ Jr.”, i.e., the suffix can be separated with a comma or not – as the author prefers to write their name.

signature. The acknowledgements signature is the text at the end of the acknowledgements and it gives the place where the author(s) is/are when writing the acknowledgements and also gives the date and name(s).

`\authorCity{A City}` specify the city

`\authorCountry{A Country}` specify the country

`\authorCityCountryDate{}` pass into this function the month and year for the acknowledgement. This can be a string such as January 2022 or it can be a $\LaTeX$ expression, such as `\MONTH\enspace\the\year`.

If there is a second author and the place, month, and year are **all** the same, then specify the month and year for only the **first** author:

`\authorCityCountryDate{\MONTH\enspace\the\year}`

If there is a second author and the place is different, then say:

`\authorCityCountryDate{}`

If there is a second author, the macros are:

`\secondAuthorsLastname{}` the last name of the 2nd author

`\secondAuthorsFirstname{}` the first name of the 2nd author

`\secondemail{}` the KTH e-mail address of the 2nd author

`\secondkthid{}` the 2nd author's kthid

`\secondAuthorsSchool{}` the school of the 2nd author

If the second author is not in the same place as the first author, then add the relevant information using the macros below. This information will be used when generating the acknowledgements signature.

`\secondAuthorCity{A City}` specify the city

`\secondAuthorCountry{A Country}` specify the country

`\secondAuthorCityCountryDate{\MONTH\enspace\the\year}`
pass into this function the month and year for the acknowledgement

If the second author is the same place as the first author, then comment out or delete the `\secondAuthorCityCountryDate{ }` as shown below:

```
%\secondAuthorCityCountryDate{ }
```

C.5 Starting to write - for the impatient

For those who are impatient and rapidly want to start writing, I suggest you start by configuring the `custom_configuration.tex` file (see Appendix C.3), your working title, and abstract (Appendix C.6.1). After this, a quick way to start writing the text in your document is to go to the table of contents in Overleaf and click on a chapter or section - this will utilize a hyperlink to go to that part of the PDF file. Next, click on the left-going arrow near the top of the border between the LaTeX on the left and the PDF on the right; this will take you (close) to the correct place in the source file where you can start to modify the content and write.

C.6 Starting to write

As you write you will notice "todo" notes in the template. They follow the following conventions:

```
\generalExpl{Comments/directions/... in English}
\sweExpl{Text på svenska}
\engExpl{English descriptions about formatting}
\sweExpl{warnings}
```

If you do not want to see these notes, you can, of course, redefine the above macros to output nothing. If you do not want to see any notes, then add the option **final** to the `\documentclass` arguments near the top of the file `examplethesis.tex`.

C.6.1 Working abstract

I generally recommend that every student start by writing a working abstract, this will help you keep your focus. To find where you can start to enter your abstract, look in the `examplethesis.tex` file for the line:

```
\generalExpl{Enter your abstract here!}
```

There is lots of information already in the template to help you with entering text, equations, *etc.*, in your abstract. **NB** Abstracts are supposed to

stand by themselves, this means no footnotes, no cross-references, no figures, no tables, *etc.*

I suggest avoiding the use of the defined acronyms in abstracts *i.e.*, spell them out rather than using the glossary commands. This is due to the fact that the `glossaries` package (that is being used to support acronyms) does not directly provide support for multiple languages and because I do not understand how to programmatically create plurals of acronyms in Swedish or other languages. Even in an English abstract, it is desirable to avoid using the glossary commands - as this makes subsequent processing of the abstracts harder - since one has to make sure that the list of acronyms and their definitions are provided to any program that will process this $\text{\LaTeX}$ source code. For this reason, later versions of this template include the `acronyms.tex` file after the metadata for DiVA.

C.6.2 Structure of the abstracts and summaries

The basic $\text{\LaTeX}$ structure for an abstract or summary is shown below (for the case of an English abstract and a Swedish summary *i.e.*, sammanfattning):

```
\begin{abstract}
  \markboth{\abstractname}{}
\begin{scontents}[store-env=lang]
eng
\end{scontents}

\begin{scontents}[store-env=abstracts,print-env=true]
here is where you abstract goes.
\end{scontents}

\subsection*{Keywords}
\begin{scontents}[store-env=keywords,print-env=true]
% If you set the EnglishKeywords earlier, you can
  ↪ retrieve them with:
\InsertKeywords{english}
% If you did not set the EnglishKeywords earlier then
  ↪ simply enter the comma separate keywords here:
%such as: Canvas Learning Management System, Docker
  ↪ containers, Performance tuning
\end{scontents}
\end{abstract}
```

```

\cleardoublepage
\babelpolyLangStart{swedish}
\begin{abstract}
  \markboth{\abstractname}{}
\begin{scontents}[store-env=lang]
swe
\end{scontents}
\begin{scontents}[store-env=abstracts,print-env=true]
Swedish summary goes here
\end{scontents}
\subsection*{Nyckelord}
\begin{scontents}[store-env=keywords,print-env=true]
% SwedishKeywords were set earlier, hence we can use
  ↪ alternative 2
\InsertKeywords{swedish}
\end{scontents}
\end{abstract}
\babelpolyLangStop{swedish}

```

It is important to note that the contents of the `scontents` environment for the abstracts are stored **verbatim**, *i.e.*, the $\text{\LaTeX}$ is **not** executed. The reason for this is to be able to later have a program that can manipulate the source $\text{\LaTeX}$ to convert it to HTML for use in announcements, calendar events, and for DiVA. This means that if you write the following:

```

\begin{scontents}[store-env=abstracts,print-env=true]
\input{abstract.txt}
\end{scontents}

```

what will end up in your abstract in the metadata save for DiVA will simply be: `"\input{abstract.tex}"` – which means that someone will have to cut and paste your actual abstract to insert it into DiVA.

It is also important to note that the following lines:

```

\begin{scontents}[store-env=lang]
eng
\end{scontents}

```

must be before the `scontents` environment for the abstracts and keywords – as these lines indicate what language the subsequent abstract and keywords are in. The three-character code used for the language is the ISO 639-2 Code – specifically the "B" (bibliographic) variant of these codes — as these codes are used in the DiVA metadata to tag what language is used.

C.6.3 Abstracts must be able to stand alone

The abstract needs to be able to stand alone; therefore, you **cannot** include citations to your references – as the references are **not** part of the abstract! It is possible (but very rare) to have footnotes as part of the abstract. However, you should be aware that quite often, if the abstract is manually entered in DiVA, the footnote might not be entered. In this case, unless your full text is available (*e.g.*, via DiVA), a reader might not have an easy way to find out what the footnote says.

C.6.4 Acronyms

You may want to define an acronym to help you with your writing, as this can both reduce the amount of typing and help your reader by providing consistent use of acronyms. The acronyms' definitions can be found in the file *lib/acronyms.tex*. The file contains some examples. I generally try to sort the lines to help find which acronyms I already have defined and keep track of the new one(s) I want to add.

C.6.5 Some predefined macros to help when writing

The file *lib/defines.tex* includes some macros that will help you when writing. This includes `\etc`, to give you “*etc.*”, `\eg`, `\ie`, and `\etal`. The file also defines `\first`, `\Second`, ... `\eighth` to give you *(i)*, *(ii)*, *(iii)*, ... *(viii)*. Note that ‘Second’ is written with an initial capital letter to avoid conflict with the unit ‘second’ in the `siunitx` package.

C.6.6 Additional abstract(s)

All theses at KTH are **required** to have an abstract in both *English* and *Swedish*. However, in addition to this, many students want to add abstracts in additional languages. The template comes pre-configured with places for abstracts in several other languages. If there is a language that you want to use that is not already supported, there are directions for how to add an additional language. If there are abstracts in languages that you do not want, please delete them or comment them out (see Appendix [C.6.7](#)).

C.6.7 Removing and hiding parts that you do not want

It is quite likely that you will find parts of the template that you do not want/need. One way of dealing with this is to delete them, and another way is to comment them out. Personally, I like to comment things out, in case I actually do want to be able to read it in the $\text{\LaTeX}$ file or uncomment it later. To comment out a portion of the file, simply use the following environment:

```
\begin{comment}
    *** what you want to comment out ***
\end{comment}
```

For example, if you are not interested in the Swedish language `todo` notes, you can look for lines with “`\sweExpl`” in them and comment them out (or delete them).

C.6.8 Removing the README_notes

At some point you will no longer want this README information. You can remove it by removing the line `\include{README_notes/README_notes}` – from the *examplethesis.tex* file. You can then remove the **README_notes** directory.

Unless you are an examiner or an administrator you can delete the file: `README_notes/README_examiner_notes.tex` and delete the include of this file from near the end of the template (*i.e.*, *examplethesis.tex*). You can also delete the directory **README_notes/README_examiner-figures**.

C.7 Copyright or Creative Commons License

It is possible to have several variants of the bookinfo page*:

copyright If you want to have a bookinfo page, include the line saying `\bookinfo`.

Creative Commons (CC) If you want to have a bookinfo page but want to have a Creative Commons license, then include `\bookinfo` and use and configure the `doclicense` package as described below.

*When printed double sided, the bookinfo page is the back of the title page.

none If you do **not** want to have a bookinfo page, comment the line saying \bookinfo page and add a \cleardoublepage.

For background about Creative Commons licenses, see: <https://www.kb.se/samverkan-och-utveckling/oppen-tillgang-och-bibsamkonsortiet/open-access-and-bibsam-consortium/open-access/creative-commons-faq-for-researchers.html> and <https://kib.ki.se/en/publish-analyse/publish-your-article-open-access/open-licence-your-publication-cc>.

Note that the lowercase version of the Creative Commons license has to be used in the modifier, *i.e.*, one of: by, by-nc, by-nd, by-nc-nd, by-sa, by-nc-sa, or zero. For the list of supported licenses, see the documentation for the doclicense package.

Note that if the doclicense package is used, it automatically redefines \bookinfo page to be \bookinfo pageCC.

C.7.1 Example configuration to have a CC BY-NC-ND license

```
\usepackage[
  type={CC},
  modifier={by-nc-nd},
  version={4.0},
  hyphenation={RaggedRight},
]{doclicense}
```

Note that the option “hyphenation=RaggedRight” can be used with the configuration of the package to set the license information with a ragged right margin rather than as a filled and justified paragraph.

C.7.2 Example configuration to have a CC BY-NC-ND license with a Euro symbol rather than a Dollar sign

```
\usepackage[
  type={CC},
  modifier={by-nc-nd},
```

```

version={4.0},
imagemodifier={-eu-88x31}, % to get Euro symbol rather
    ↪ than Dollar sign
hyphenation={RaggedRight},
]{doclicense}

```

C.7.3 Example configuration to have a CC0 license

```

\usepackage[
    type={CC},
    modifier={zero},
    version={1.0},
]{doclicense}

```

C.8 Use of fonts within the thesis

The choice of fonts is a very individual matter and may be affected by the kind of content that you are trying to write, the language that you are writing in, and what you want to convey to your reader. However, some points to keep in mind are:

- Use fonts with serifs for the body of your thesis, their presence makes it much easier for your reader.
- Use sans serif fonts for headings. This helps your reader distinguish them from the body.
- Be very careful when using fonts that are not widely available*. Unless you embed the fonts that you have used, your readers may not see what you want them to see. Ideally, you should embed all fonts – even if you only embed the subset you use.
- Although there are fonts that have a huge number of characters in them, they might not have the characters that you need.

*For example, even though it is widely used, not everyone has the Arial font. Additionally, it is a proprietary font; thus, you need to have an appropriate license to use it.

- There are also fonts that, although they have a vast number of characters in them, do not have the math table that $\text{\LaTeX}$ needs to be able to set mathematical content^{*}.
- Many fonts are proprietary, thus you need to consider whether you have an appropriate license to use them.

What can you do when the fonts you use are missing characters that you need to use? One solution is to use a font that has the character(s) that you want and then make use of them in the places that you need to.

The details of working with different fonts and characters is a rather complex area and not for the faint-hearted. However, if you **really** want to have specific characters, $\text{\XeLaTeX}$ and $\text{\LuaLaTeX}$ have the means to help you realize what you want.

C.9 One big thesis file or a master file with includes of the parts

While many students split their thesis into multiple files (such as `introduction.tex`, `background.tex`, `method.tex`, `what-you-did.tex`, `results-and-analysis.tex`, `discussion.tex`, `conclusion-and-future-work.tex`) and then include these in their main document (with a series of `\includexxxx`), my experience is that it actually makes it hard to be consistent in the thesis. For example, you cannot do a simple global replacement when you have decided to introduce a particular acronym. It also makes searching for things difficult, as Overleaf's search function only works on individual files[†]. Personally, I find it hard to correct LaTeX errors when the file is split in this way - since Overleaf does not make it simple to find the root cause of a problem when the chapters are included in this way. There can be a problem with compiling the project in Overleaf, as Overleaf does not always handle the separate files as one document (unless you use the functions to tell LaTeX that a file is part of a larger document and identify the parent document). Although I have had some students do this splitting successfully and they liked being able to compile just

^{*}An example of such a font is Google's Noto font. Even though it includes a vast number of characters, it lacks a math table – although there is an awareness of this missing feature

[†]Note that this is not a problem if you use emacs and a tags file, as this can do searching over the whole set of files and even a tags based query and replace.

a part of their report; I've personally had strange errors occur with it - hence I did not use this with the template.

There are some advantages to splitting the document into different parts:

- Overleaf has a limit on the number of changes that it can track - but this limit is per file! [Yes, I have gotten bitten by this when I have put in more changes and comments than the limit and had to stop marking up a manuscript.]
- Additionally, Overleaf has a per file size limit (*i.e.*, how large a file can be) - again this is per file [Yes, I have gotten bitten by this when exporting a Jupyter notebook that produced a LaTeX file larger than 50 MB.]
- This is useful when different students (in a 1st cycle degree project) are writing different parts of the report (in this case the divisions can be even at the section or subsection level).

Similarly, many students like to group their figures along with their chapters, *i.e.*, introducing a folder for each part of the report and placing both the text and the figures relevant to this section into the relevant folder. A similar approach can be used with included code snippets, tables, *etc.*

Ultimately, I think the main issue is the degree to which the separate files are separate and can be worked on as if they were very independent. This generally is true in third-cycle theses, as the chapters tend to be rather independent - typically with one conference/journal paper as the focus of a chapter. However, my experience is that first-cycle theses have very highly interdependent parts, while 2nd cycle theses are split between highly interdependent and highly independent.

However, some might find the question of splitting or not to be a matter of taste or perhaps different ways to approach organizing their writing. So you and your supervisor(s) and examiners might want to discuss what choice is most suitable for your purposes.

Appendix D

README and notes about the template

This document, written by Gerald Q. Maguire Jr, describes the thesis template that I have developed for use at **KTH Royal Institute of Technology (KTH)** and provides some background about why it is the way that it is. It is important to note that the template is **not prescriptive**, as not every thesis will have all the parts the template shows. However, if there is something that you decide to leave out, you should make a conscious decision to do so, and you should consider the impact this may have on your thesis being approved by the examiner.

Fundamental to the design of the template are several key factors:

- Helping students be successful in their degree project,
- Helping students produce a high-quality thesis, and
- Supporting all of the (relevant) phases of the degree project process.

Several thousand theses are written each year by **KTH** students. Every approved thesis will be entered into **Digitala Vetenskapliga Arkivet (DiVA)** (independent of whether the full text is made available via **DiVA**). Collecting the data necessary for **DiVA** was a major driving force in the design of the template. This data is useful for many of the phases of the degree project, such as announcing the oral presentation.

This template is **not** designed for use by **Interactive Media Technology (TIMTM)** and **Media Management (TMMTM)** students - as students in these two programs are using a different structure for their reports (there is another template available for them).

This document is a work in progress.

D.1 Introduction

This template evolved (radically) from an earlier thesis template that was widely used at **KTH**. The direction of this evolution was based on the DOCX template developed over many years for use with students for whom I was the examiner and/or supervisor. The suggested structure and contents of the thesis reflect my experience as an examiner for more than 600 degree projects and the experience I have had as a teacher and examiner for the course *II2202 Research Methodology and Scientific Writing*. The template also reflects my interest as a member of KTH's Language Committee in facilitating the parallel use of English and Swedish at **KTH**, as well as supporting other languages. The latter aspect reflects my experience with double-degree students, who often need to have at least the abstract of their thesis in their home university's language(s). The thesis template also reflects my experience in entering the metadata for hundreds of theses into **DiVA** and announcing a very large number of degree project seminars.

Appendix **D.4** describes several different groups of users and how the template is relevant to them.

Several major thoughts have influenced the design of this template:

- Thought 1** The template should help a student be successful in their degree project and help them produce a high-quality thesis in conjunction with their degree project.
- Thought 2** The template should help support all of the (relevant) phases of the degree project process.
- Thought 3** Redundant data entry should be minimized to increase consistency.
- Thought 4** There are several thousand theses written each year at **KTH**. Theses are the second most common type of publication at **KTH**.
- Thought 5** Every approved thesis will have at least its metadata entered into **DiVA**. **DiVA** features multi-language support for title, subtitle, abstract, and keywords.

D.2 Delimitations

This template is **not** designed for use by **TIMTM** and Media Management **TMMTM** students - as students in these two programs are using a different

structure for their reports (there is another template available for them).

Additionally, I have been told by one of my colleagues in applied mathematics that theses in this area generally do not follow the **Introduction, Methods, Results, and Discussion (IMRAD)** structure.

Some parts of the template are conditional based on the value of a switch: `\ifinswedish`. The idea is to easily have a single template that supports theses written in English or Swedish. However, in many places, the conditional has not been used but could be. Examples of this include the Swedish names for chapters and sections. Generally, this information is in a note after the English chapter or section name. More complete implementation of the use of this condition remains as future work.

The template does not fully support the G5 paper format. In particular, the **KTH** cover (produced with `\kthcover`) and back cover (produced with `\kthbackcover`) have only been adapted for A4 paper. Support for G5 paper remains as future work.

The handling of the subject area (Swedish: Område för examensarbete) is currently incomplete and remains as future work. Personally, I'm still struggling to understand the rules and how one knows what the correct values are (especially for cases of *(i)* dual degrees and *(ii)* combinations of technical subjects and education degrees).

D.3 Structure of the files for the template

Table **D.1** shows the structure of the files for the template. These files are generally taken from an existing Overleaf project, a ZIP file, or a github.

One hope is that by automatically extracting information from various sources, this information is more likely to be *correct* and *consistent* (supporting **Thought 3**). This approach has been used to generate two of the files used for the template. These files are:

1. The file `custom_configuration.tex` contains macros and values for configuring a project. These values are generally expected to be known at the start of the project, *e.g.*, author(s), supervisor(s), examiner, course code for the degree project program code, *etc.* While this file can be manually edited, it was designed to be generated by a program that I have written that extracts most of the data from the Canvas course being used in conjunction with the degree project. One of the goals of using such a program is to extract data from Canvas automatically, the **KTH** profile **Application Programming Interface (API)**, **Kurs- och**

programplaneringssystemet (KOPPS), and other sources. The macros for defining this information are described in Sections **C.4**, **D.5.1**, and **D.5.2** - for authors, supervisors, and examiner (respectively).

2. The file `schools_and_programs.ins` contains the English and Swedish names of schools and programs. A program extracted this information from **KOPPS**.

We will assume that these files have been generated by someone. Later we will examine who this someone might be for each of these files.

Table D.1: Structure of files for the template

bibstyle	directory containing files related to the style of the bibliography	
	<code>myIEEEtran.bst</code>	a bibtex style file
figures	directory containing files for figures	
lib	directory containing various library files	
	<code>acronyms.tex</code>	a place to define the acronyms that will or might be used
	<code>defines.tex</code>	some generally useful defines
	<code>includes-after-hyperref.tex</code>	a special include file for packages that have to be included after the hyperref package
	<code>includes.tex</code>	a centralized place to include packages that might be useful
	<code>kthcolors.tex</code>	defines a number of colors from the KTH palette
	<code>pdf_related_includes.tex</code>	includes to be able to add the title and other information to the PDF file
	<code>schools_and_programs.ins</code>	English and Swedish names of schools and the programs
<code>custom_configuration.tex</code>	macros and values for configuring a project	
<code>examplethesis.tex</code>	an example of the thesis itself	
<code>kth_logo.png</code>	the KTH logo for use on the cover	
<code>KTH_ROYAL_INSTITUTE_OF_TECHNOLOGY_logotype.png</code>	KTH logotype for use on the English language cover	
<code>kththesis.cls</code>	the kththesis class file	
<code>README_notes.tex</code>	these notes	
<code>references.bib</code>	references that may be cited in the thesis	

D.4 Expected users and their differences

This template is relevant to several different sets of users:

- Users 1** Author or Authors (see Appendix C.1),
- Users 2** Those working together with the author(s) during the degree project process (see Appendix D.5),
- Users 3** Administrative staff working with the document after it has been approved by the examiner (see Appendix D.6), and
- Users 4** The (hopefully) many (human) readers of the final document (see Appendix D.7).
- Users 5** The (hopefully) many computers reading the metadata and the full text of the final document (see Appendix D.7.1).
- Users 6** Those who are maintaining or updating this template (see Appendix D.7.2).

Each of these different sets of users has different needs and perspectives. The following subsections describe these needs and perspectives.

For information for authors, see Appendix C - located in the file `README_author.tex`.

D.5 Those working in parallel with the authors(s) during the degree project

Those working together with the author(s) during the degree project process include the examiner, supervisor(s), and the opponent(s).

D.5.1 Supervisor

If a degree project is done in industry, there is generally an industrial supervisor in addition to the academic supervisor(s). The template supports up to 5 supervisors (typically an academic supervisor, an industrial supervisor, and sometimes an additional academic or industrial supervisor). The choice of up to five reflects my experience, observation of prior theses in DiVA, and a request by a student with four supervisors. Note that there is expected to be at least one supervisor. The supervisors are enumerated as A, B, C, D, and E.

For each of A, B, C, D, and E as appropriate, replace the "X" in the following macros:

`\supervisorXsLastname{ }` the last name of the supervisor
`\supervisorXsFirstname{ }` the first name of the supervisor
`\supervisorXsEmail{ }` e-mail address of the supervisor

If the supervisor is from within **KTH**, then add their KTHID, School, and Department info:

`\supervisorXsKTHID{ }` the supervisor's kthid
`\supervisorXsSchool{ }` the school of the supervisor
`\supervisorXsDepartment{ }` the department of the supervisor

If the supervisor is from outside of **KTH**, then add their organization with:

`\supervisorXsOrganization{ }` the supervisor's organization

D.5.2 Examiner

I assume that there is only a single examiner for a given thesis*. For this examiner, the relevant macros are:

`\examinersLastname{ }` the last name of the examiner
`\examinersFirstname{ }` the first name of the examiner
`\examinersEmail{ }` e-mail address of the examiner

If the examiner is from within **KTH**, then add their KTHID, School, and Department info:

`\examinersKTHID{ }` the examiner's kthid
`\examinersSchool{ }` the school of the examiner

*Statistically, there are very few theses with multiple examiners, and this generally occurs for students either in a double degree program or when there are two students in a 1st cycle degree project from different schools, then there might be one examiner for each student. As the case of more than one examiner occurs very infrequently, I have left it for future work. The pseudo-JSON structure is set up to handle multiple examiners, but additional macros would be needed in a similar fashion as used for multiple supervisors, and this metadata would have to be conditionally added where appropriate.

`\examinersDepartment{ }` the department of the examiner

If the examiner is from outside of **KTH**, then add their organization with:

`\examinersOrganization{ }` the examiner's organization

I assume that someone (such as the examiner) will generate the file: `custom_configuration.tex`. This assumption is based upon the fact that the examiner knows who the student or students are who will be working on a given degree project, who the supervisor or supervisors are, what program the student is in, course code, Ideally, this file should be generated automatically by some computer program so that each student or pair of students in a group gets a customized template automatically via the Canvas course. However, currently, the file is generated using a command line program (`create_customized_JSON_file.py`) to generate a **JavaScript Object Notation (JSON)** file. Subsequently, a separate program (`customize_LaTeX_project.py`) takes this **JSON** data and creates the appropriate $\text{\LaTeX}$ commands and inserts this information into the file and then inserts this file into a ZIP file, either replacing or augmenting the `custom_configuration.tex` within this ZIP file (if one exists). There is an option for this second program `-initialize` that causes the program to simply replace the file rather than appending the new information to the end of the file.

The above programs are available from <https://github.com/gqmaquirejr/E-learning>. The README file for this GitHub contains information about how to run the programs, their options, and gives examples.

D.5.3 Opponent(s) and oral presentation

Unlike the supervisors and examiner, the macros related to the opponent and oral presentation are in the `examplethesis.tex` file. The macro for the opponent(s) is:

`\opponentsNames{ }` the names (in normal name order) of
the opponent or opponents

When there are multiple opponents, separate their names with `'\&'`; for example, A. B. Normal `\&` A. X. E. Normal.

For the oral presentation, the following macros are filled in once the examiner has scheduled your oral presentation:

<code>\presentationDateAndTimeISO{}</code>	date and time of the presentation is ISO format, for example: 2022-03-15 13:00
<code>\presentationLanguage{}</code>	three letter abbreviation for the language of the presentation according to three letter ISO 639-2 Code – specifically the "B" (bibliographic) variant of these codes (note that this is the same language code used in DiVA), generally eng or swe
<code>\presentationRoom{}</code>	a room name and/or "via Zoom https://kth-se.zoom.us/j/ddddddddd "
<code>\presentationAddress{}</code>	location of the room, for example: Isafjordsgatan 22 (Kistagången 16)
<code>\presentationCity{}</code>	city where the presentation occurs, generally: Stockholm

D.6 Administrative staff

Once a thesis is approved by the examiner we need to add the TRITA number. The TRITA number is assigned by the student affairs office of the school from an annual series of numbers.

D.6.1 What is a TRITA number and why does each approved thesis get assigned one?

TRITA stands for Transactions for the Royal Institute of Technology, with the letter 'A' appended to it. The TRITA definition is the 1971 report, "Mall för publikationsserier vid Kungl. Tekniska högskolan i Stockholm", TRITA-LIB-1001, <http://urn.kb.se/resolve?urn=urn:nbn:se:kth:diva-127656>.

The format for TRITA numbers for degree projects is TRITA-⟨school acronym⟩-EX-YYYY:nnnn, where nnnn is a sequential number starting from 1 each year with the numbers assigned in chronological order to approved theses (“numren delas ut kronologiskt först när examinatorn godkänt arbetet.” - according to one of KTH’s archivists). Note that the list of assigned TRITA numbers is archived each year*. The year, YYYY, is based on the year that the thesis was *approved*.

The TRITA number value can be set with a macro that takes two arguments: series and year:number as shown below:

```
% for entering the TRITA number for a thesis
\trita{TRITA-XXX-EX}{2022:00}
```

D.6.2 Where does the TRITA number go?

The TRITA number will appear on the back cover of the thesis. It is also stored as part of the metadata entered into **DiVA**.

D.6.3 What does this mean in practice?

Currently, at School of XXX[†] the TRITA number is only assigned to the thesis when the examiner has approved the thesis and submitted the PDF of the approved thesis (with cover) to the student affairs office. Of course, this does not make much sense because the back cover is already on the thesis! This means that someone in the student affairs office must either (i) edit the sequential number part of the TRITA number (using some PDF tool) or (ii) they need to make a new back cover and replace the existing back cover. A better solution would be to inform the examiner of the TRITA number and the examiner can see that this number is inserted into the macro shown above and this can enable the number to appear on the back cover and as an added bonus be included in the metadata for **DiVA**.

Note that it is expected that in 2023, this process will change – thus the assignment of the TRITA number and the application of the back cover would be done by the student affairs office (as only they have the relevant information)[‡].

*It seems that this archiving is done twice a year.

[†]Name replaced to neutralize the template.

[‡]Note to maintainers: This means that the back cover can be removed from this template.

D.6.4 Entering the metadata into DiVA

If a thesis has used this template the “For DIVA” page contains the metadata for **DiVA** and an administrator can cut and paste this data into **DiVA**. Alternatively, this metadata can be extracted with a program from the PDF file to produce a **JSON** file that can subsequently be used to create a MODS file for import into **DiVA**. The $\text{\LaTeX}$ compiler can in many cases produce a file called “fordiva.json” that contains the metadata.

The programs that can be used to extract data and to take a **JSON** file and create a MODS file are available from <https://github.com/gqmagui/rejr/E-learning>.

Note that the import of the MODS file does **not import the collaboration data**, even though this is in the file. This is a limitation of the **DiVA** import function. Therefore, this information has to be manually entered along with uploading of the PDF file itself.

D.7 (Human) Readers of the thesis

Some theses have very few downloads from **DiVA**, while some have had hundreds of thousands of downloads. Therefore, you should remember that you have a wide range of human readers of your thesis. The readers include other students looking for information related to their own thesis or because they are interested in the future work that you have suggested to work on for their own degree project. Additionally, researchers who are looking for your results may find your thesis relevant to them. In many cases, companies will look at theses for ideas about what the state of the art is - in several cases, theses have been important as “prior art” and this invalidated patents that had been issued if the patent was submitted after the thesis became public (hence it pays to get theses public as soon as possible). Other human readers are the UKÄ review teams that examine the degree programs offered at **KTH**. Finally, as **KTH** is a public agency, it is important that the general public know what is done at **KTH**^{*}.

D.7.1 Machines reading the metadata or full text of the thesis

The file `pdf_related_includes.tex` contains $\text{\LaTeX}$ code that stores the title, author(s), and keyword information into the PDF document in such

^{*}This is an important part of the Swedish Offentlighetsprincipen.

a way that if you ask for the properties of the PDF file you will get this data. This information makes it easier for machines to get this information from the PDF file.

Additionally, many search engines (such as Google’s search engine) mine **DiVA** for the metadata and if the full text of the thesis is published via **DiVA** then they also process the full text of the thesis. The result is that search engines can find the content in these theses. This is likely to increase the probability that someone will download your thesis if they think it is relevant to them – increasing the number of your human readers (see Appendix **D.7**).

D.7.2 Template author and maintainers

KTH periodically changes the cover design for theses, introduces new programs of study, eliminates programs of study, reorganizes administratively, and faculty move between schools, departments, and divisions. It can be expected that this template will need to evolve with these changes.

For example, if there is a change in schools or programs then there needs to be changes made to the file `schools_and_programs.ins`. While the current file was extracted from **KOPPS**, the program that does this will need to be replaced because further development of **KOPPS** has been terminated by KTH’s central IT unit which plans to transition all of this information to **Lokalt adb-baserat dokumentationssystem (LADOK)**.

As another example, on 13 December 2021 there was a change in the **KTH** cover for 1st and 2nd theses, and the cover generator web service was shutdown. The initial draft version of the cover used a proprietary font (TheSans B4 SemiLight and TheSans B6 SemiBold). The version that was publicly introduced uses another proprietary font (Arial) and officially only existed as a DOCX file for a thesis in Swedish. The result is that I had to make my own version in $\text{\LaTeX}$ to try to emulate the DOCX cover. This led to a lot of effort, but one can get a reasonable cover with the correct font as described in Appendix **C.2**.

D.8 More about Abstracts

In addition to abstracts of the form shown in the template, there is another form of abstracts, called structured abstracts. These are described in Appendix **D.8.1**. Some automated checking of abstracts has been added to the template, see Appendix **D.8.2**.

D.8.1 Structured abstracts

In some fields, structured abstracts are commonly used. These typically have the following parts:

Background/Objectives

`<text>`

Method

`<text>`

Results

`<text>`

Conclusions

`<text>`

Such structured abstracts are typically longer (in word count) than typical abstracts. For an example of such a structured abstract, see [28]. The document is available in both English (*Humor as a protective factor against anxiety and depression*) and Spanish (*El humor como factor protector de la ansiedad y la depresión*) via <https://pmc.ncbi.nlm.nih.gov/articles/PMC6994741/>.

D.8.2 Automated checking of some properties of abstracts

To avoid common mistakes, the template now checks the contents of the abstracts and outputs warnings if the author puts a citation or cross-reference into the abstract.

The method of checking was inspired by the code used in the ACM Conference Proceedings templates to give a warning if the user uses a `\vspace` command or redefines `\baselinestretch`, see Listing D.1. To check the abstract, a pair of functions was introduced to save and restore commands, as was done in the package `phfnote`, see Listing D.2.

Listing D.1: Code snippet from ACM Conference Proceedings template

```

\let\@vspace@orig=\@vspace
\let\@vspacer@orig=\@vspacer
\apptocmd{\@vspace}{\ClassWarning{\@classname}{\string\
  ↪ vspace\space should
  only be used to provide space above/below surrounding
  objects}}{}{}
\apptocmd{\@vspacer}{\ClassWarning{\@classname}{\string\
  ↪ vspace\space should
  only be used to provide space above/below surrounding
  objects}}{}{}
\let\@vspace@acm=\@vspace
\let\@vspacer@acm=\@vspacer
\let\ACM@origbaselinestretch\baselinestretch
\AtEndDocument{\ifx\baselinestretch\ACM@origbaselinestretch\
  ↪ else
  \ClassError{\@classname}{An attempt to redefine
  \string\baselinestretch\space detected. Please do not do
  ↪ this for
  ACM submissions!}\fi}

```

Listing D.2: Code snippet from phfnote package

```

%% Macros from the package phfnote
% \phfnoteSaveDefs<{}>{identifier}<{}>{list of macro >names}
  ↳ saves the current definitions of the given list of
  ↳ macros under the identifier. The list of macros is
  ↳ specified as a comma-separated list of macro names.

\def\phfnoteSaveDefs#1#2{%
  \csgdef{phfnote@restoreddefs@#1}{}%
  \def\@tmpa{#2}%
  \for\next:=\@tmpa\do{%
    \global\csletcs{phfnote@restoreddefs@#1@\next}{\next}%
    \expandafter\xappto\csname phfnote@restoreddefs@#1\
      ↳ endcsname{%
      ↳ \noexpand\csletcs{\next}{phfnote@restoreddefs@#1@\next}
      ↳ %
    }%
  }%
}

% \phfnoteRestoreDef<{}>{identifier} - restores the saved
  ↳ macros
\def\phfnoteRestoreDefs#1{%
  \ifcsname phfnote@restoreddefs@#1\endcsname%
    \csname phfnote@restoreddefs@#1\endcsname%
  \else%
    \PackageError{phfnote}{\string\phfnoteRestoreDefs: no
      ↳ such
      ↳ definitions stored (#1)}{}
  \fi%
}

```

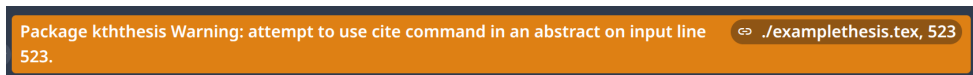
Given these two new macros, we use hooks (provided by the `etoolbox` package) to save the definition of `\cite` before the abstract and restore it afterwards – as well as to redefine `\cite` to produce a warning if used within the abstract. Note that in the PDF output, there will be **!?cite!?** at the place (in the PDF) where the `\cite` was - to draw the user's attention to the error. The same mechanism is used for several other macros. The code for these hooks is shown in Listing D.3.

Listing D.3: Hooks to save and restore the cite and ref definitions before and after the abstract environment

```
\BeforeBeginEnvironment{abstract}{\phfnoteSaveDefs{
  ↪ origcmds}{cite, ref, Cref, cref}%
\renewcommand{\cite}[1]{\{\NotoSansFont \textcolor{red}{!}{\}
  ↪ \uuline{cite!}{\}}\PackageWarning{kththesis}{attempt
  ↪ to use cite command in an abstract}}%
\renewcommand{\ref}[1]{\{\NotoSansFont \textcolor{red}{!}{\}
  ↪ \uuline{ref!}{\}}\PackageWarning{kththesis}{attempt
  ↪ to use ref command in an abstract}}%
\renewcommand{\Cref}[1]{\{\NotoSansFont \textcolor{red}{!}{\}
  ↪ \uuline{Cref!}{\}}\PackageWarning{kththesis}{
  ↪ attempt to use Cref command in an abstract}}%
\renewcommand{\cref}[1]{\{\NotoSansFont \textcolor{red}{!}{\}
  ↪ \uuline{cref!}{\}}\PackageWarning{kththesis}{
  ↪ attempt to use cref command in an abstract}}%
}

\AfterEndEnvironment{abstract}{\phfnoteRestoreDefs{
  ↪ origcmds}}
```

The hooks shown in Listing D.3 produce a warning message that provides the page number where the `\cite` was used in an abstract (see Figure D.1). This code could be expanded to handle other undesirable commands in an abstract.



```
Package kththesis Warning: attempt to use cite command in an abstract on input line 523.
./examplethesis.tex, 523
```

Figure D.1: Example of a warning for use of a citation in an abstract)

The combination of two red Exclamation Question Marks (!?, unicode U+2049) with the command between them (double underlined) was selected as: (i) this string is unlikely in the user’s text, (ii) it was colored red to make it highly visible, and (iii) the offending command is double underlined to help make it visible (even to those who cannot see the color red). While this combination of characters is sometimes turned into an Interrobang glyph (‽, unicode U+203D), I have chosen to use the Exclamation Question Mark glyph as I found the Interrobang too subtle visually. Additionally, when using struck-through or struck-out characters rather than the double underline, it was too hard to read the name of the command. While the PDF does not include the arguments to the `\cite` or `\ref` command - these are in the source $\text{\LaTeX}$.

D.8.3 Tables and images in abstracts

This template and the tools that I have written to help enter the metadata into DiVA do **not** support tables and figures in abstracts.

As an abstract in DiVA can contain HTML code, it is technically possible to make tables and include figures in a DiVA abstract. The problems are (i) that I do not know a general means of transforming a table from LaTeX to HTML, and (ii) I do not know of a good way of passing the image (or images) that might be in a LaTeX abstract (via tools) to DiVA. The second problem might be addressed by attaching the image files to the PDF and adding code to the generated HTML to refer to these images. One problem with this is that the images would probably need to be uploaded as supplemental/supporting material, and then there would need to be some tool to update the abstract to refer to this uploaded file.

In a recent correction that I suggested for a thesis abstract in DiVA, I manually created the table in HTML to match the table that was in the abstract in the printed thesis.

The DiVA administrator who entered the thesis into DiVA had not put *any* of the abstract in, but simply put "See file" in the abstract field!

D.8.4 Graphical or visual abstracts

Karolinska Institutet (KI) has an excellent page about graphical abstracts, see [Graphical abstracts](#). At present, this template does **not** provide any special support for graphical abstracts. One can use the usual mechanisms to include a figure in the abstract; however, see Appendix [D.8.3](#).

I am unsure (i) how to deal with such figures with respect to facilitating their inclusion in DiVA (Could it be uploaded separately as supplemental material?), and (ii) it is unclear to me whether this graphical abstract would be placed before or after the keywords in the LaTeX file. This will require some further evaluation.

Recently, I helped create such a graphical abstract (for a journal paper) using `tikz`. This might have some advantages in terms of accessibility if the `tikz` source code were available to a (screen) reader.

D.9 Cover illustration

Some theses feature an optional cover illustration. Although I have not gotten an answer from the KTH communications unit about the specifications of the cover illustration, I have tried to infer something suitable from a DOCX file.

Due to the lack of a specification for the cover illustration, one should treat this feature of my template as an experimental feature that will be subject to change. Use at your own risk.

Two macros are introduced to make this process simpler, see Listing D.4. The first (`\coverIllustration`) provides the filename of the illustration to be used. The template includes an `image1.png` file. This blue rectangular image was taken from the DOCX template. The author should replace this file with the illustration that they want to use.

NB There are some constraints about what you can put on the cover. If in doubt, query KTH's [Communications Department \(COM\)](#).

The second macro (`\coverIllustrationCredit`) enables the author to acknowledge the creator of the illustration. Here, we will assume that the author has the creator's permission to include the illustration on the cover of their thesis. If the author has created their own illustration, then they do not need to specify any credit, as it is implicitly the work of the author.

NB The author may have to manually adjust the height and width options to the `\includegraphics` command in the `kththesis.cls` file to get the results that they want. If you make the figure too large, it will appear on a second page and **not** on the cover. If this occurs, you need to reduce the height of the illustration.

Listing D.4: Configuring a cover illustration

```
% It is possible to have a cover illustration
\coverIllustration{figures/image1.png}
% if so, the author might acknowledge this or give the
  ↳ copyright information for it with:
\coverIllustrationCredit{A. B. Normal}
```

D.10 While writing

As was noted in Appendix C.1, the thesis template contains lots of examples, notes, and comments. One method to provide additional information is the use

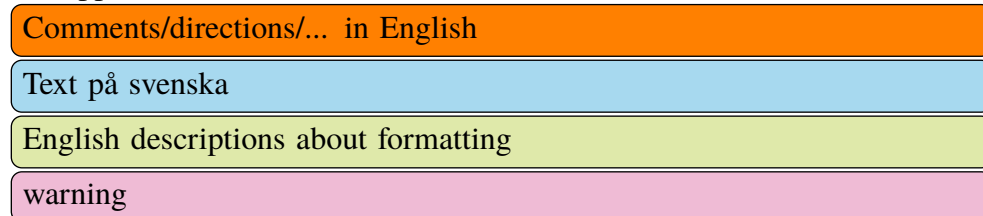
of `\todo`. Several different types of `\todo` notes have been used in the thesis. These are described in Appendix [D.10.1](#).

D.10.1 Conventions for `\todo` notes

The example thesis text includes extensive comments, directions, and warnings. These follow the form shown below:

```
\generalExpl{Comments/directions/... in English}
\sweExpl{Text på svenska}
\engExpl{English descriptions about formatting}
\warningExpl{warning}
```

and appear as:



Each of the above is a macro, so as usual in $\text{\LaTeX}$ you can redefine it - even defining it to produce nothing! Several previous students have placed these re-definitions in the `custom_configuration.tex` file.

D.10.2 Turning on and off the `README_notes`

As the various `README` notes are targeted at different readers, you may or not want to see them. It is very easy to turn them on or off by adding or removing a percent (`'\%`') character before the relevant `\begin{comment}` and `\end{comment}` comments around each set of notes.

For example, if you are a student writing a thesis, I suggest turning off everything except for the `README_author.tex` and `README_notes.tex` sets of notes. However, I would suggest keeping the other `README` files around (at least for a little while) as a source of examples of how to do things. Despite having spent a very large number of hours working on the template and drafts of students' theses, I find some of the `README` files very helpful as a reminder of how to do things.

D.10.3 Removing the `README_notes`

At some point you will no longer want this `README` information. You can remove it by removing the line `\include{README_notes/README_notes}`

– from the `examplethesis.tex` file. If you have removed the other README* files from the **README_notes** directory, you can then remove the **README_notes** directory.

D.10.4 Removing the README_notes

At some point, you will no longer want this README information. You can remove it by removing the line `\include{README_notes/README_notes}` – from the `examplethesis.tex` file. If you have removed the other README* files from the **README_notes** directory, you can then remove the **README_notes** directory.

D.10.5 Removing unused fonts

This version of the template may also have some font information, in the form of OpenType Font files (with the extension “.otf”) and TrueType Font font files (with the extension “.ttf”). If you are not using these fonts (and no longer are using any of the README files), then you can delete these font files.

README acronyms

API	Application Programming Interface
DiVA	Digitala Vetenskapliga Arkivet
IMRAD	Introduction, Methods, Results, and Discussion
JSON	JavaScript Object Notation
KI	Karolinska Institutet
KOPPS	Kurs- och programplaneringssystemet
KTH	KTH Royal Institute of Technology
LADOK	Lokalt adb-baserat dokumentationssystem
TIMTM	Interactive Media Technology
TMMTM	Media Management

€€€€ For DIVA €€€€

```
{
  "Author1": { "Last name": "Kourangi Esfahani",
    "First name": "Arian",
    "Local User Id": "u1azvhtx",
    "E-mail": "arianke@kth.se",
    "organisation": { "L1": "School of Electrical Engineering and Computer Science",
    }
  },
  "Cycle": "2",
  "Course code": "DA236X",
  "Credits": "30.0",
  "Degree1": { "Educational program": "Master's Programme, Systems, Control and Robotics, 120 credits"
    , "programcode": "TSCRM"
    , "Degree": "Both Degree of Master of Science in Engineering and Master's degree"
    , "subjectArea": "Electrical Engineering"
  },
  "Title": {
    "Main title": "Tetherless Collaborative Transportation in Space via Velocity-Matched Pushing",
    "Subtitle": "Utilizing STL to enable gentle transport of objects in space using collaborative untethered agents",
    "Language": "eng"
  },
  "Alternative title": {
    "Main title": "Kollaborativ Transport i rymden via hastighets-matchad knuff",
    "Subtitle": "Användning av STI för att möjliggöra varsam transport av object i rymden via kollaborativa okopplade agenter",
    "Language": "swe"
  },
  },
  "Supervisor1": { "Last name": "Verhagen",
    "First name": "Joris",
    "Local User Id": "u1qiuoh5",
    "E-mail": "jorisv@kth.se",
    "organisation": { "L1": "School of Electrical Engineering and Computer Science",
    "L2": "RPL"
    }
  },
  "Examiner1": { "Last name": "Tumova",
    "First name": "Jana",
    "Local User Id": "u19qofjk",
    "E-mail": "tumova@kth.se",
    "organisation": { "L1": "School of Electrical Engineering and Computer Science",
    "L2": "RPL"
    }
  },
  },
  "National Subject Categories": "10201, 10207, 20202, 20201",
  "SDGs": "9",
  "Other information": { "Year": "2026", "Number of pages": "xvii,100"},
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  "Abstract[eng]": €€€€

\generalExpl{Enter your abstract here!}
An abstract is (typically) about 250 and 350 words (1/2 A4-page) with the following components:
% key parts of the abstract
\begin{itemize}
  \item What is the topic area? (optional) Introduces the subject area for the project.
  \item Short problem statement
  \item Why was this problem worth a Bachelor's/'Masters thesis project? (\ie, why is the problem
    both significant and of a suitable degree of difficulty for a Bachelor's/'Masters thesis project?
    Why has no one else solved it yet?)
  \item How did you solve the problem? What was your method/insight?
  \item Results/Conclusions/Consequences/Impact: What are your key results/\linebreak[4]conclusions?
    What will others do based on your results? What can be done now that you have finished - that could
    not be done before your thesis project was completed?
\end{itemize}

"Abstract[swe]": €€€€

\generalExpl{Enter your Swedish abstract or summary here!}

€€€€,
"Keywords[eng]": €€€€
KeywordA, KeywordB, KeywordC €€€€,
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"Keywords[swe]": €€€€
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NyckelordA, NyckelordB, NyckelordC €€€,
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acronyms.tex

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%%% Local Variables:
%%% mode: latex
%%% TeX-master: t
%%% End:
% The following command is used with glossaries-extra
\setabbreviationstyle[acronym]{long-short}
% The form of the entries in this file is \newacronym[label]{acronym}{phrase}
% or \newacronym[options]{label}{acronym}{phrase}
% see "User Manual for glossaries.sty" for the details about the options, one example is shown below
% note the specification of the long form plural in the line below
\newacronym[shortplural={MIQP}, longplural={Mixed Integer Quadratic Programming}]{MIQP}{MIQP}{Mixed
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